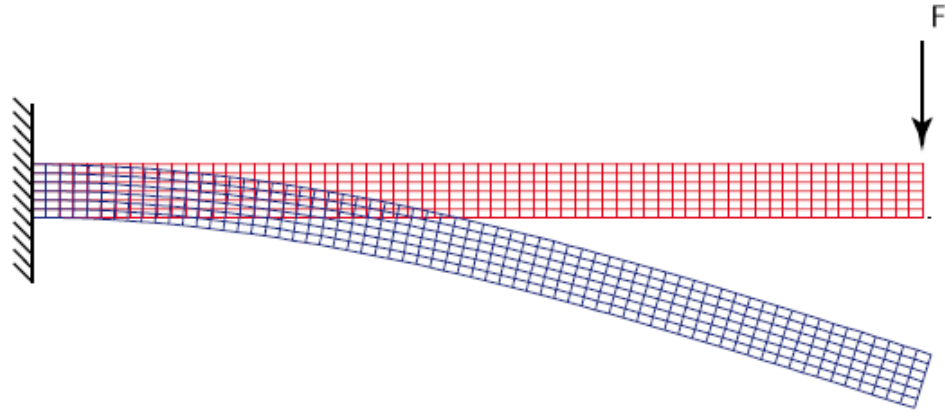
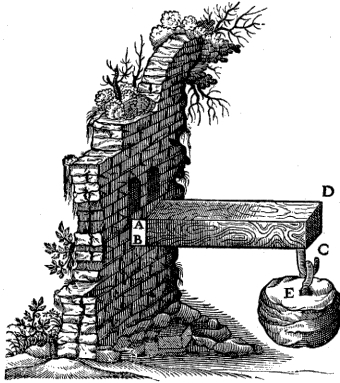


Method of Finite Elements I



Lecturers: Prof. Dr. Eleni Chatzi, Dr. Adrian Egger

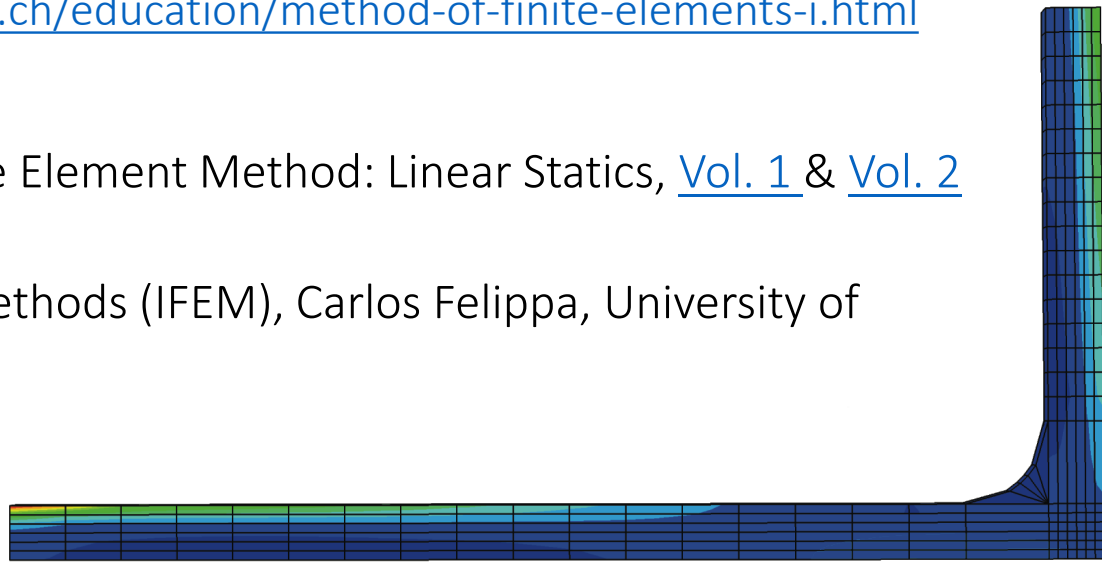
Assistants: Konstantinos Vlachas (HIE 19.5)

Homepage: <http://www.chatzi.ibk.ethz.ch/education/method-of-finite-elements-i.html>

Course books

[O] Structural Analysis with the Finite Element Method: Linear Statics, [Vol. 1](#) & [Vol. 2](#) by Eugenio Onate

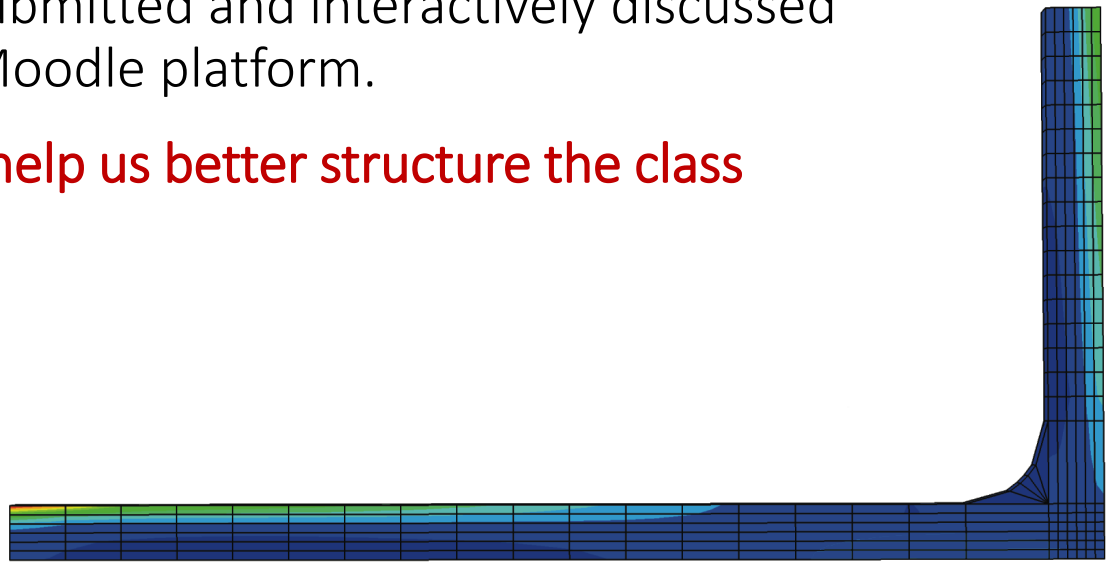
[F] Introduction To Finite Element Methods (IFEM), Carlos Felippa, University of Colorado at Boulder, [available online](#)



Method of Finite Elements I

Course Organization

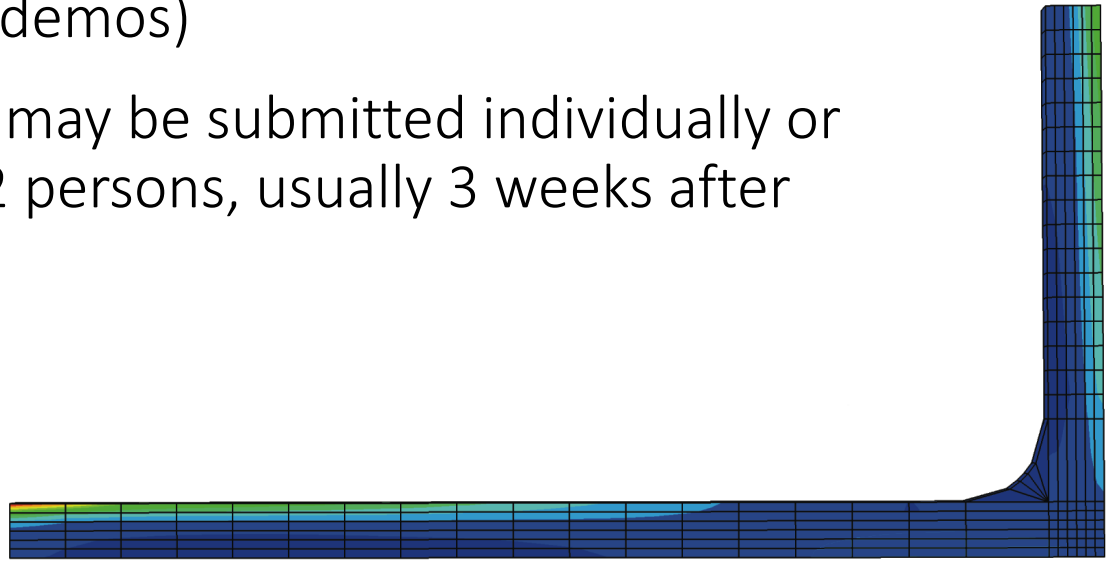
- Lectures posted weekly on the class [webpage](#) (last year's lecture slides also made available)
- Demos posted on the [Moodle platform](#) (access via your ETH login)
- Visiting hours: sign up via Moodle platform.
- Further questions can be submitted and interactively discussed via the Q&A forum of the Moodle platform.
- **Please ask questions! They help us better structure the class**



Method of Finite Elements I

Course Evaluation

- The final grade is calculated as
 - ✓ 45% from 3 **graded Homeworks** (15% each) and
 - ✓ 55% by an end of semester examination, offered on the last week of the course
- The Homeworks will require coding in Python (we offer an intro and support during demos)
- Homework Assignments may be submitted individually or by a group of maximum 2 persons, usually 3 weeks after their announcement.



Method of Finite Elements I

The Finite Element Method (FEM) is a main enabler for engineering simulations

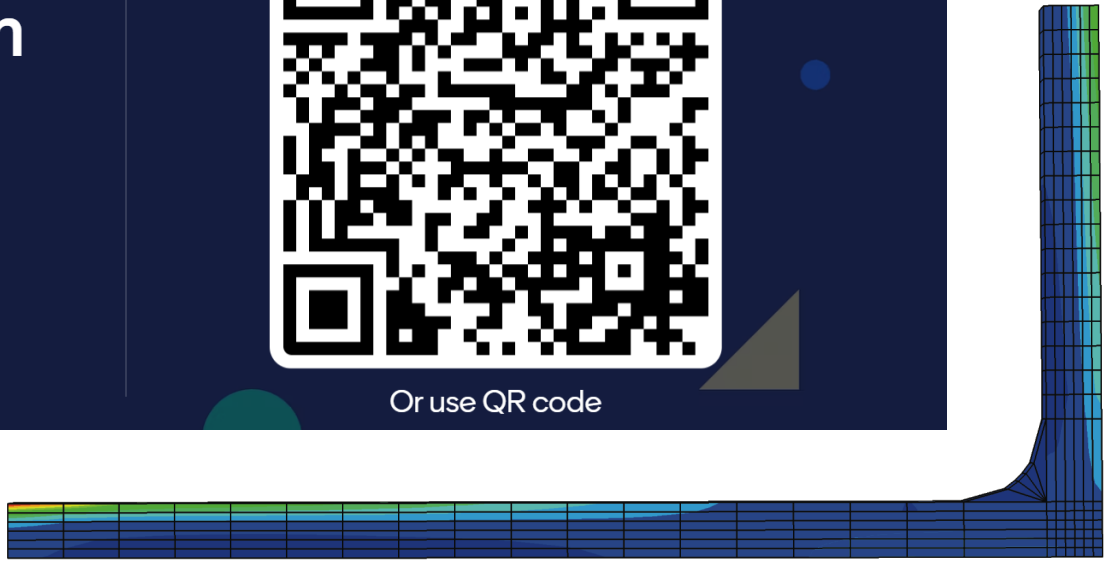
Why Use Simulation?



Go to
www.menti.com
Enter the code
5375 8470

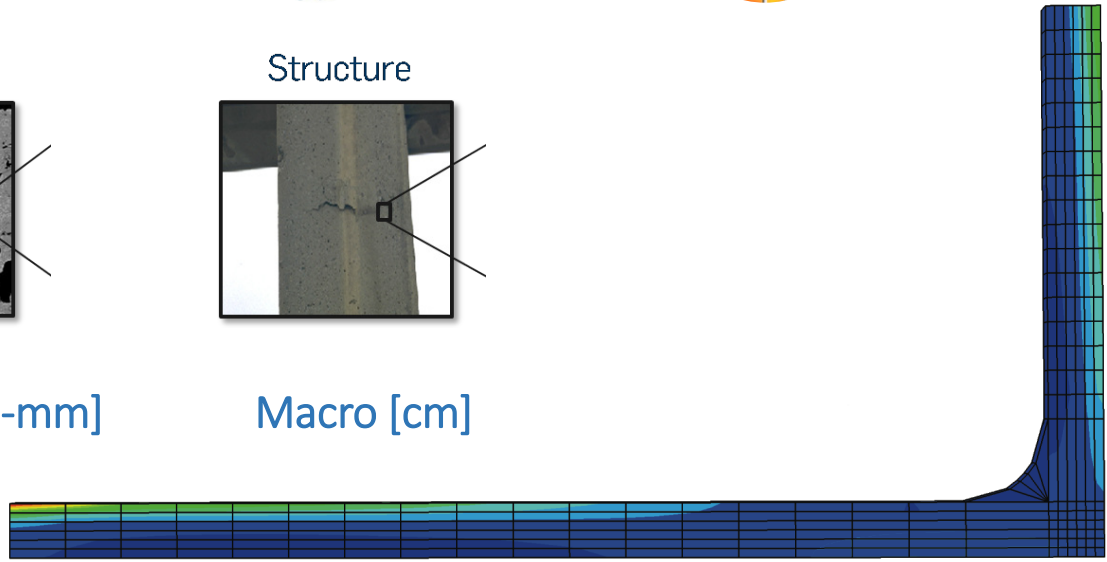
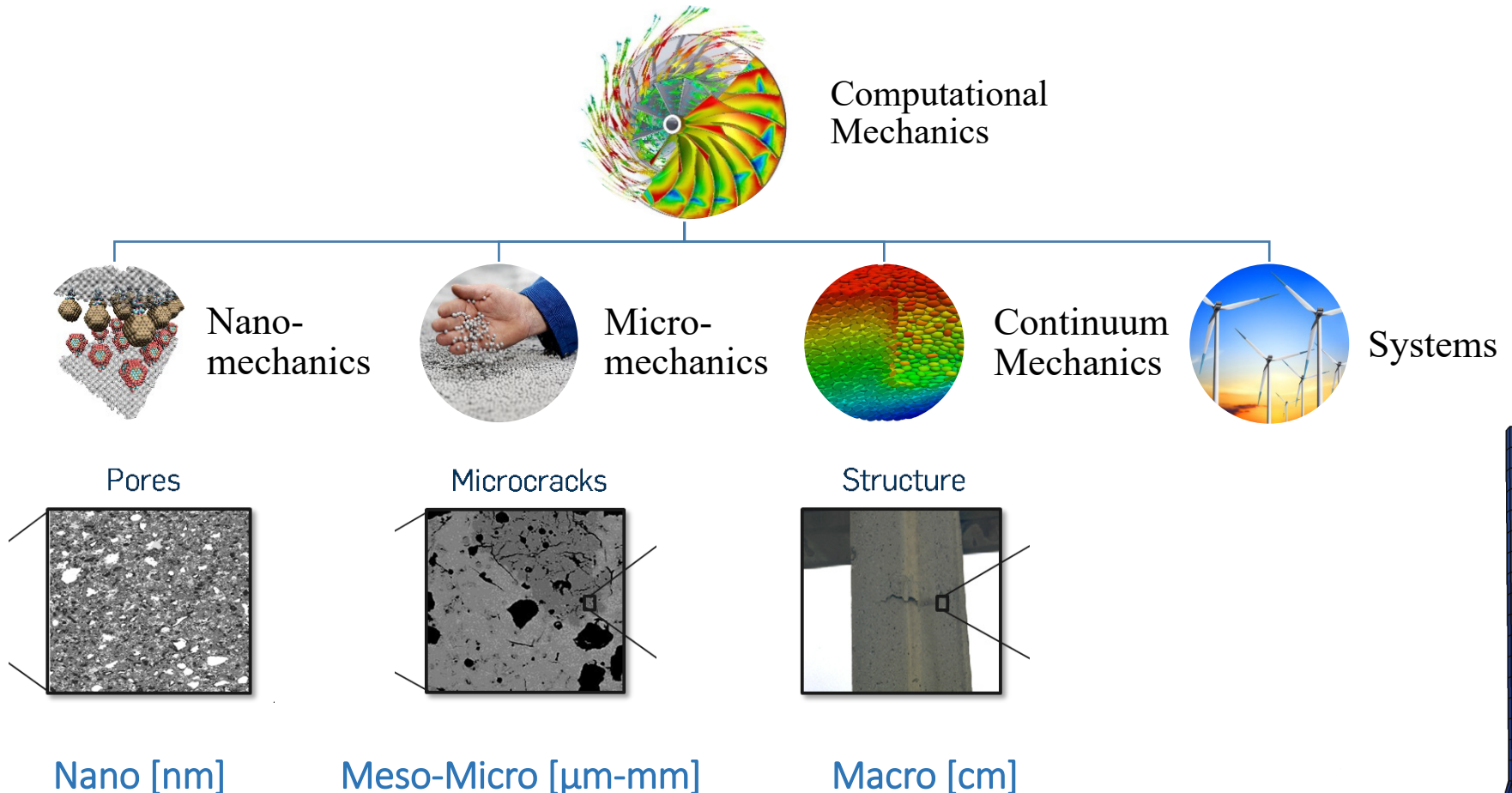
Or use QR code

The image shows a dark blue banner for a Menti quiz. On the left, it says 'Go to www.menti.com', 'Enter the code', and '5375 8470'. On the right, there is a large QR code and the text 'Or use QR code'. The banner has decorative geometric shapes in purple, blue, and grey.



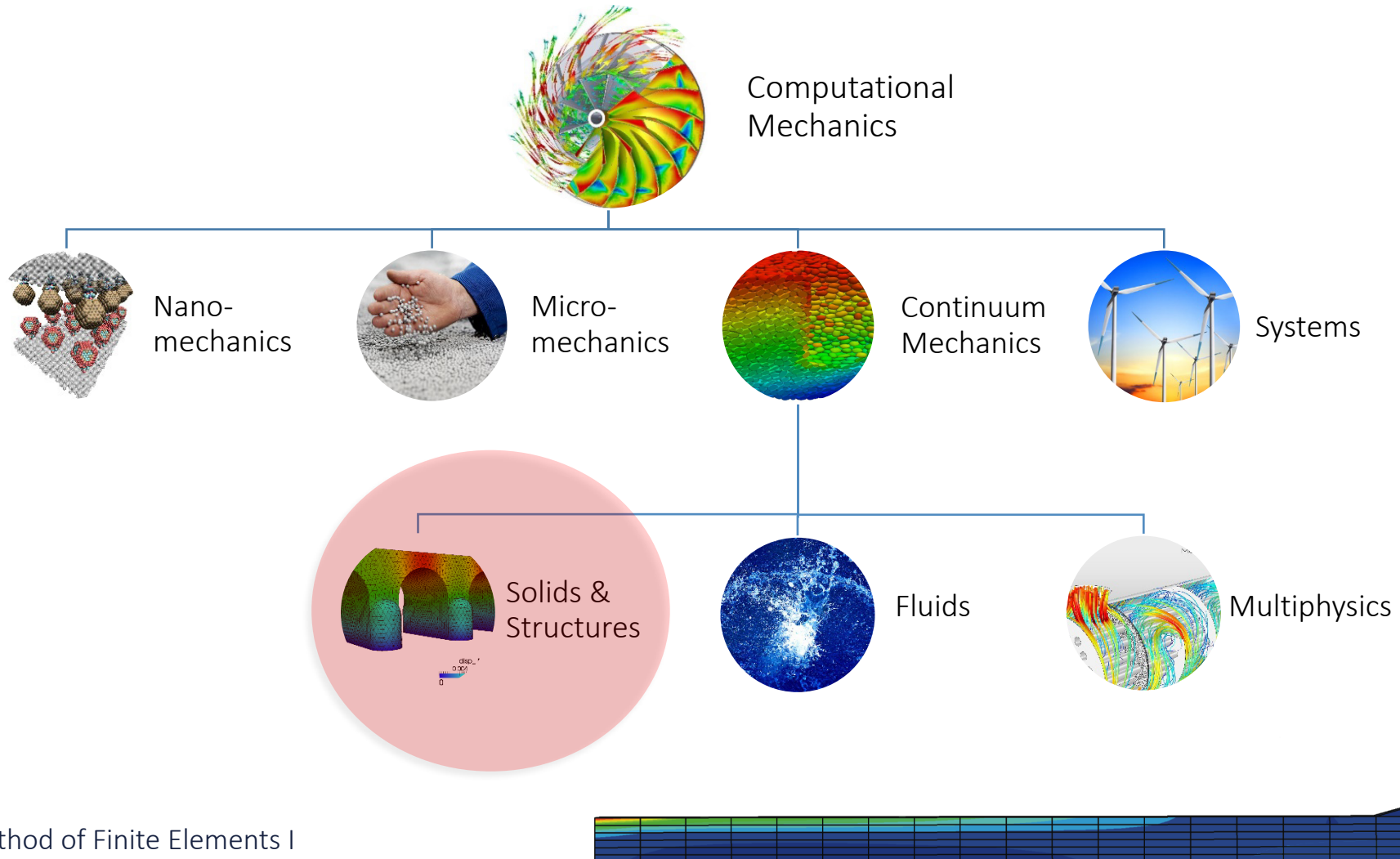
Method of Finite Elements I

Fields of Computational Mechanics



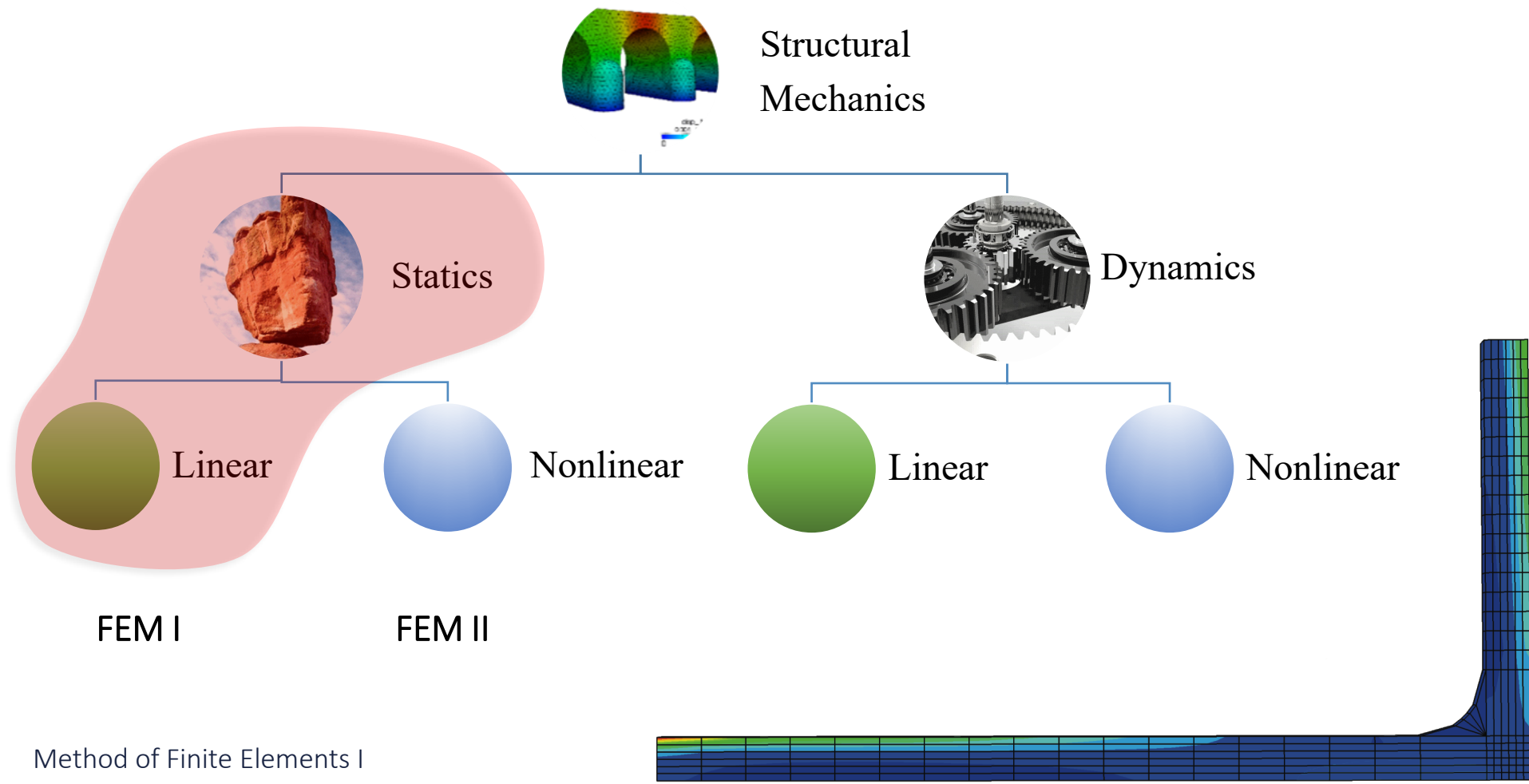
Method of Finite Elements I

Structural Mechanics



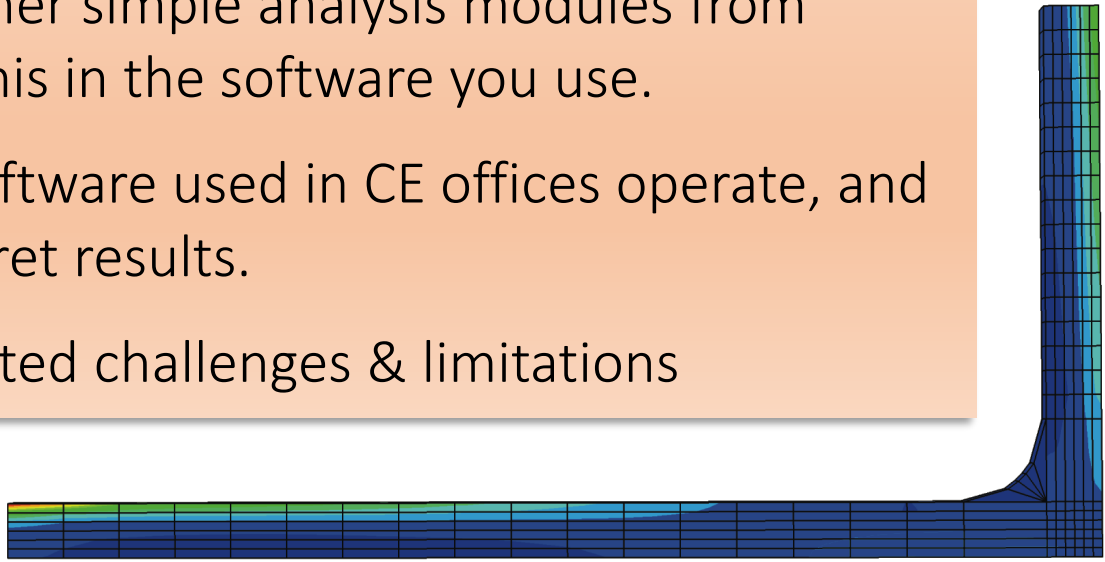
Method of Finite Elements I

Structural Mechanics Focus of this Class



Goals of the Class

- To present the linear FEM from a structural engineering perspective
- To introduce the concept of discrete computation and link this to the analytical approach taught during the Bachelors (Baustatik).
- To help you grasp the fundamental theory, and recognize how it is put into practice
- To learn how to put together simple analysis modules from scratch - then recognise this in the software you use.
- To understand how the software used in CE offices operate, and understand how to interpret results.
- To understand the associated challenges & limitations



Goals of the Class

Developing trust in simulations

©Eric Kuusisto

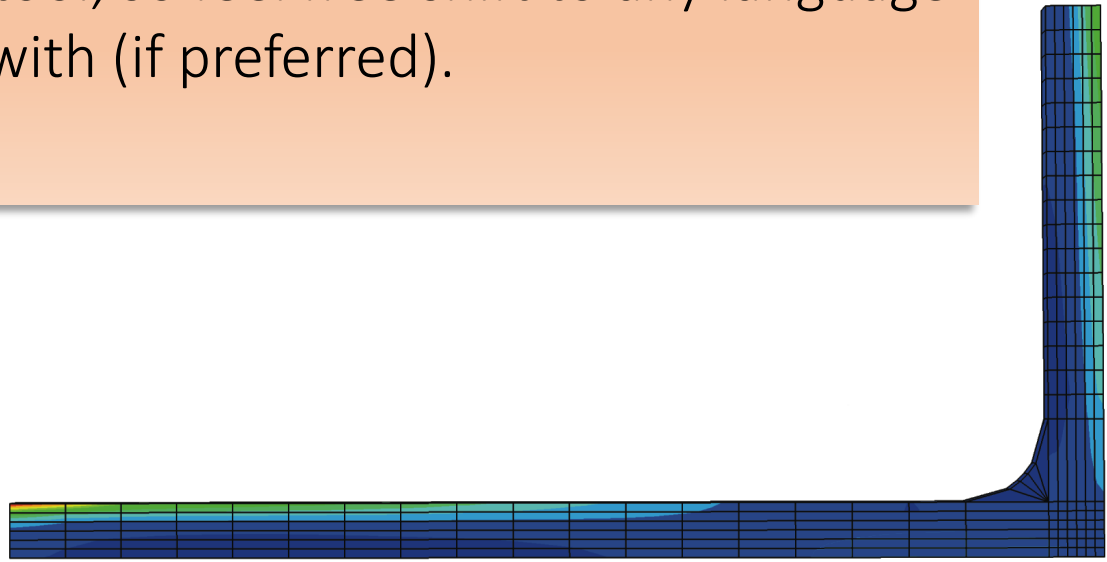


Goals of the Class

- Remember! Software is continually changing but the core of the methods remains unchanged.
- It is the fundamental core that you need to master!

PYTHON will be used as a tool in this path to learning.

It is however nothing but a tool, so feel free shift to any language you feel more comfortable with (if preferred).



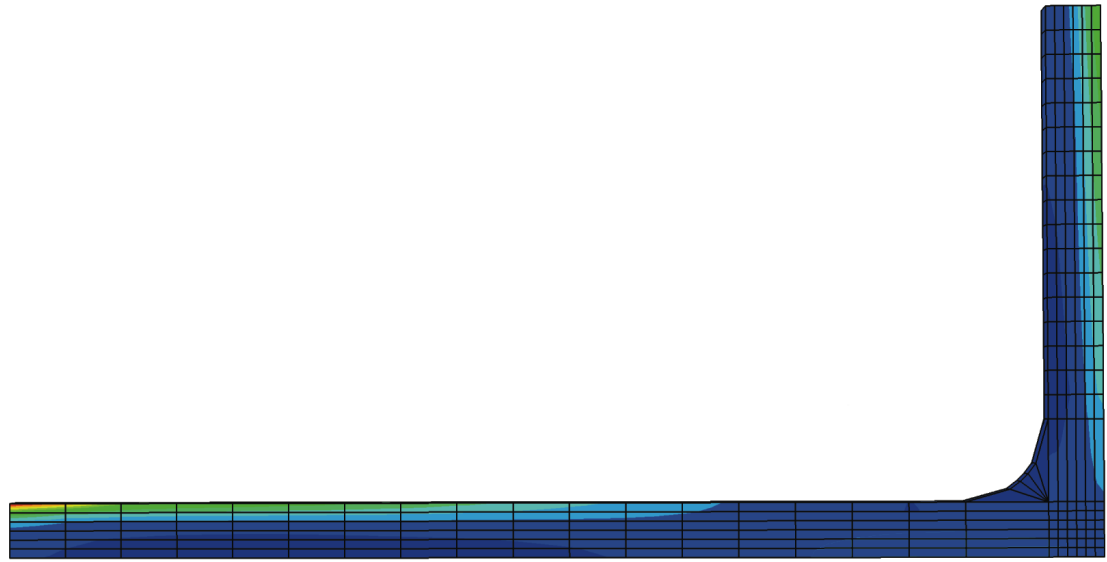
Chapter 1

Introduction to the MFE

Book Chapters

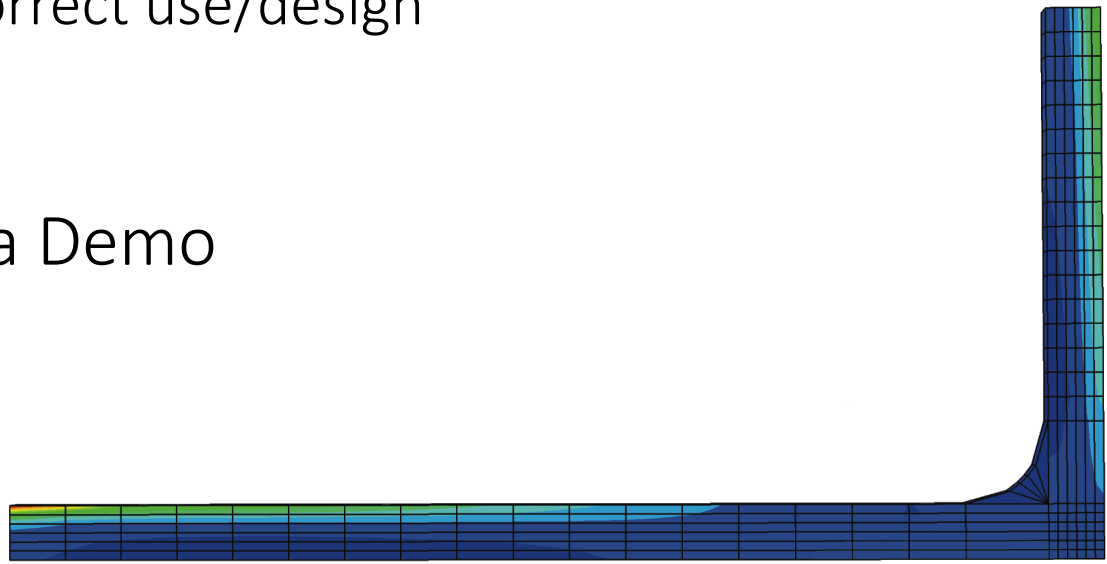
[O] V1/Ch1/1-13

[F] Ch1



Today's lecture

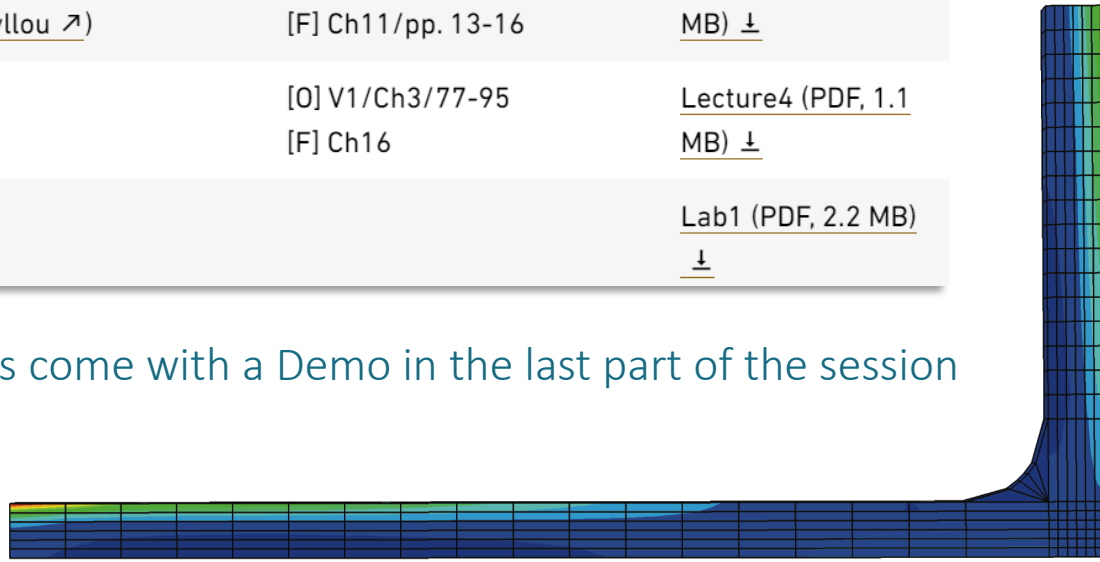
- An overview of the MFE I course
- MFE: A historical overview
- Introduction to the use of Finite Elements
 - Why and when to use FEM?
 - Challenges for a correct use/design
- FEM into Practice
- Kicking it off with a Demo



Course Overview

Date	Subject(s)	Book Chapters	Lecture
19.02	Chapter 1: Introduction to the MFE Demo: a short intro to Python	[O] V1/Ch1/1-13 [F] Ch1	Lecture1 (PDF, 6 MB) ↓
26.02	Chapter 2 - Part a: The Direct Stiffness Method Demo: Plane Truss Analysis	[O] V1/Ch1/114-34 [F] Ch2, Ch3	Lecture2a (PDF, 1 MB) ↓
04.03	Chapter 2 - Part b: Stability & Modal Analysis Demo: Plane Frame Modal Analysis	[F] Ch31, Ch32	Lecture2b (PDF, 1.5 MB) ↓
11.03	Chapter 3: The Variational Formulation (slides in collaboration with Dr. S. Triantafyllou ↗)	[O] V1/Ch2 [F] Ch11/pp. 13-16	Lecture3 (PDF, 1.7 MB) ↓
18.03	Chapter 4: The Isoparametric Formulation Demo: Demonstration on Bar Element	[O] V1/Ch3/77-95 [F] Ch16	Lecture4 (PDF, 1.1 MB) ↓
25.03	Computer Lab Quadrature & Boundary Conditions		Lab1 (PDF, 2.2 MB) ↓

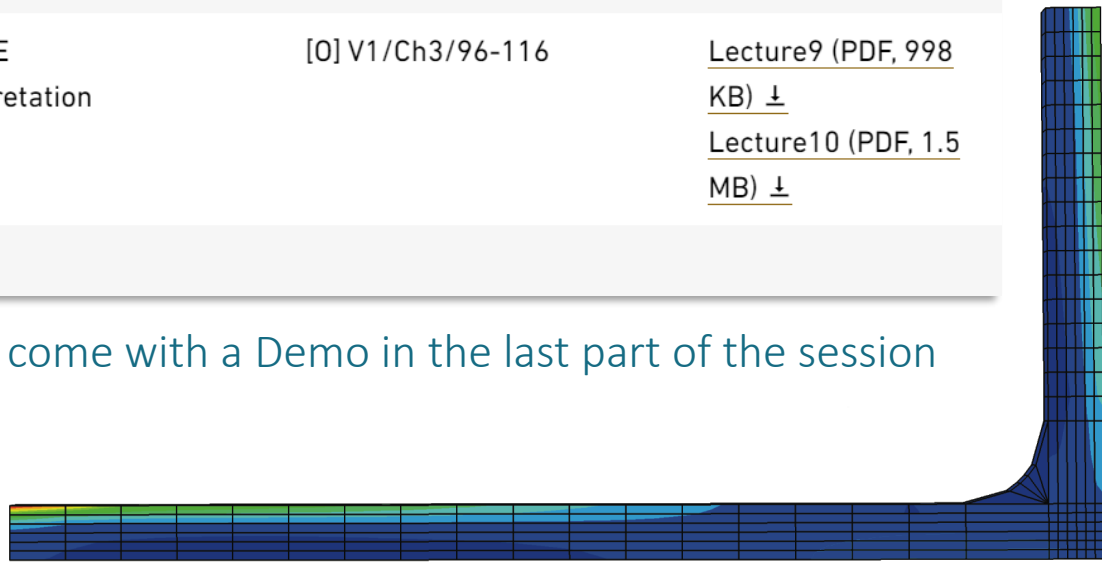
*lecture sessions come with a Demo in the last part of the session



Course Overview

25.03	Computer Lab Quadrature & Boundary Conditions		Lab1 (PDF, 2.2 MB) ↓
08.04	Chapter 5: The Euler Bernoulli Beam (slides in collaboration with Dr. S. Triantafyllou ↗) Demo: Extension to 3D	[O] V2/Ch1	Lecture5 (PDF, 2.2 MB) ↓
22.04	Chapter 6: The Timoshenko Beam Demo: Euler vs Timoshenko	[O] V2/Ch2 [F] Ch13	Lecture6 (PDF, 633 KB) ↓
29.04	Chapter 7: 2D Elements Demo: Multipoint Constraints	[O] V1/Ch4, Ch6	Lecture7 (PDF, 1.5 MB) ↓
06.05	Chapter 8: The Plate Element Demo: Plane-stress and plate elements	[O] V2/Ch5, Ch6	Lecture8 (PDF, 1.5 MB) ↓
13.05	Chapter 9: Practical application of the MFE Practical Considerations & Results Interpretation	[O] V1/Ch3/96-116	Lecture9 (PDF, 998 KB) ↓ Lecture10 (PDF, 1.5 MB) ↓
27.05	Final Exam		

*lecture sessions come with a Demo in the last part of the session



Chapter 1: FE Analysis in brief...

The goal of Structural Analysis: Modeling a system by means of approximation.

How to do this?

Come up with a simplified idealization. This may oftentimes be achieved conceptually.

Example

©Martin St-Amant



Chapter 1: FE Analysis in brief...

The goal of Structural Analysis: Modeling a system by means of approximation.

How to do this?

Come up with a simplified idealization. This may be achieved conceptually.

Example of a conceptual model

How to model this bridge?

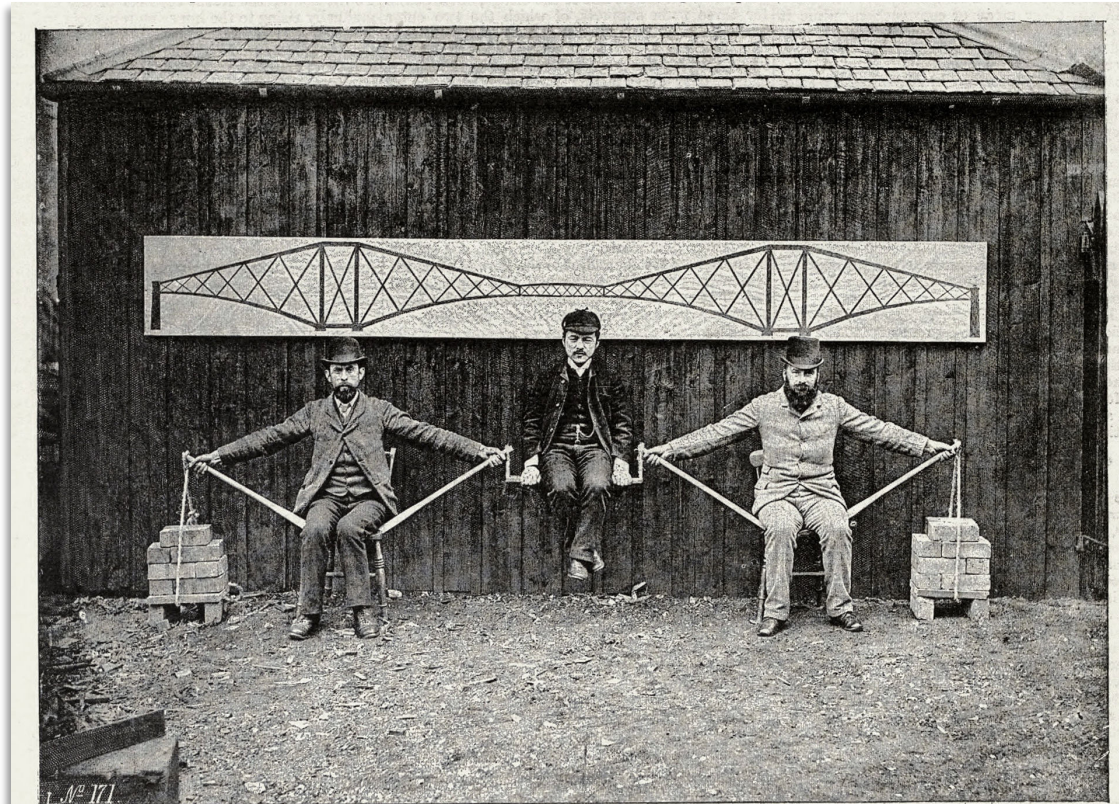


FIG. 5A. LIVING MODEL ILLUSTRATING PRINCIPLE OF THE FORTH BRIDGE.

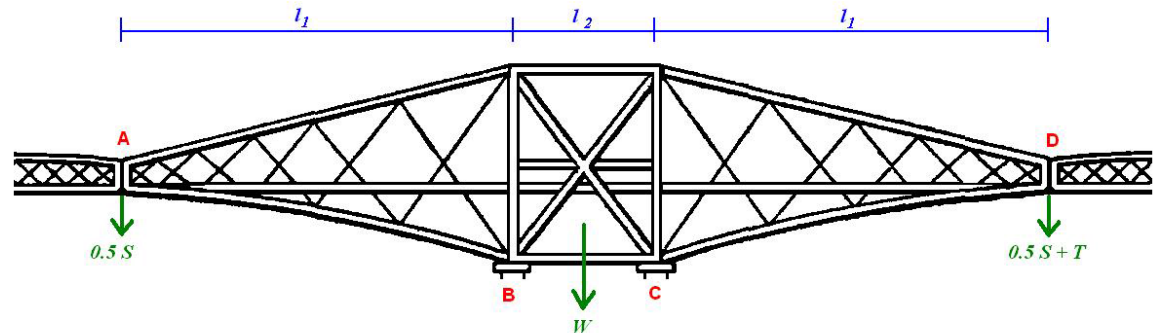
Chapter 1: FE Analysis in brief...

The goal of Structural Analysis: Modeling a system by means of approximation.

After selecting a conceptual model of a structure, the next step for the numerical (and analytical) study is the definition of a structural model (sometimes called mathematical model). These models could be of different complexity

Example

How to model this?



[©A.D. Magee: A CRITICAL ANALYSIS OF THE FORTH BRIDGE](#)

Option 1: Break the system down to 1D components, such as truss, beams elements and supports.

Can we still model this by hand? Perhaps...

Chapter 1: FE Analysis in brief...

The goal of Structural Analysis: Modeling a system by means of approximation.

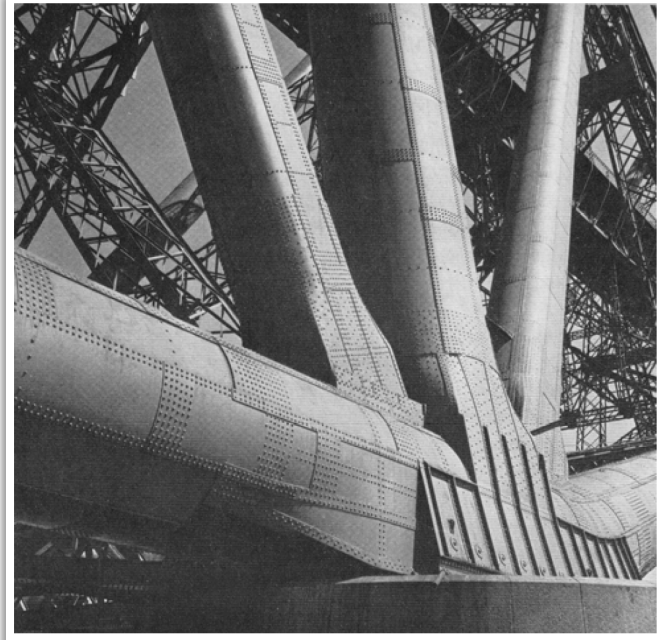
Different structural models could be set up using different complexity in our geometric and mathematical representation.

Example

What if we want to model a critical component of this system?

Can we still model this by hand?

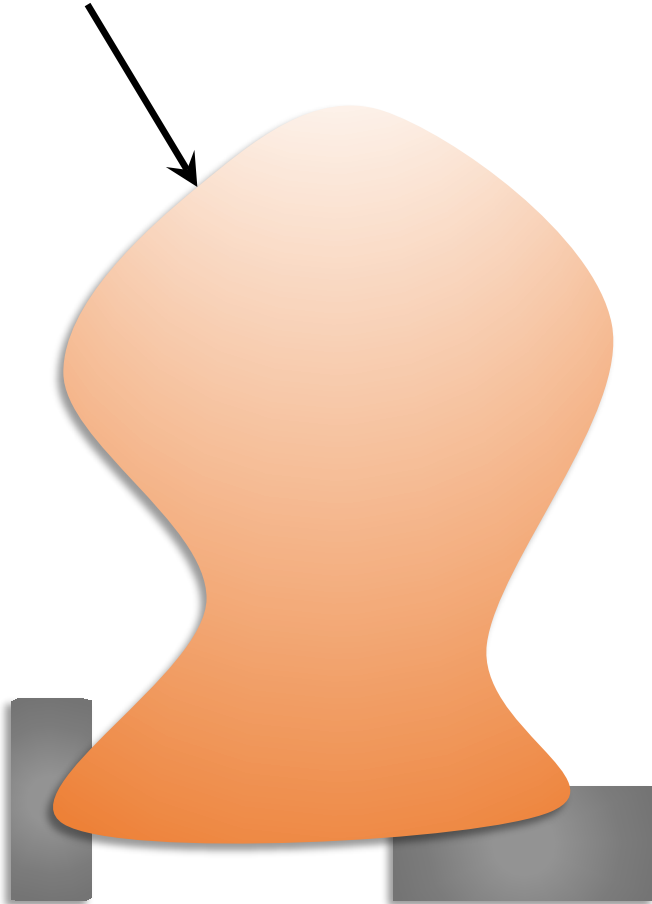
The task is now harder as it requires 2D or 3D effects accounted for



[©A.D. Magee: A CRITICAL ANALYSIS OF THE FORTH BRIDGE](#)

Chapter 1: FE Analysis in brief...

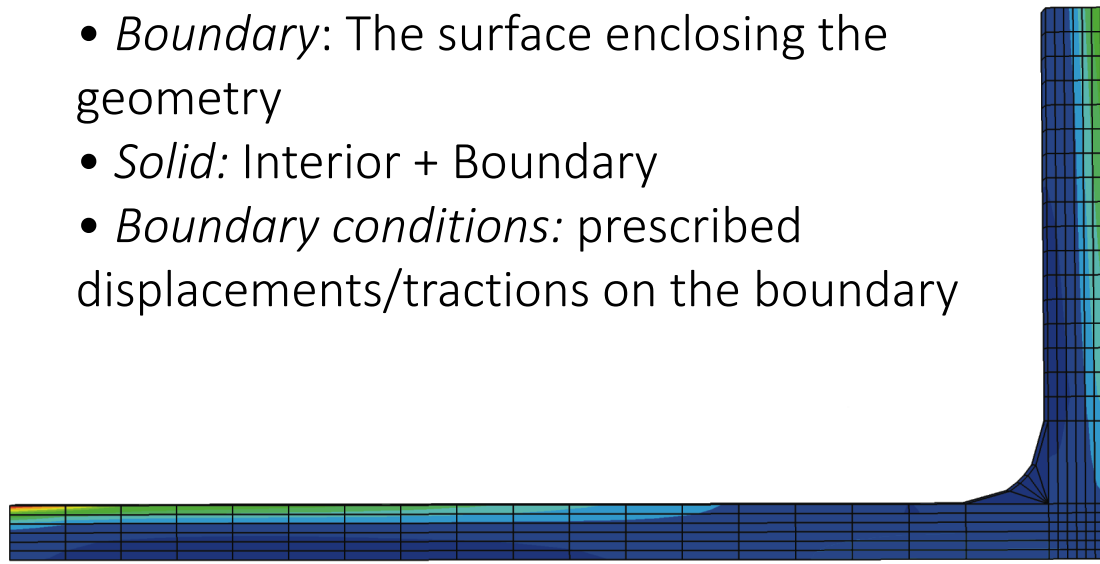
FEA was originally developed for solid mechanics applications.



Object: A Solid with known mechanical properties. (a skyscraper; a shaft; bio tissue ...)

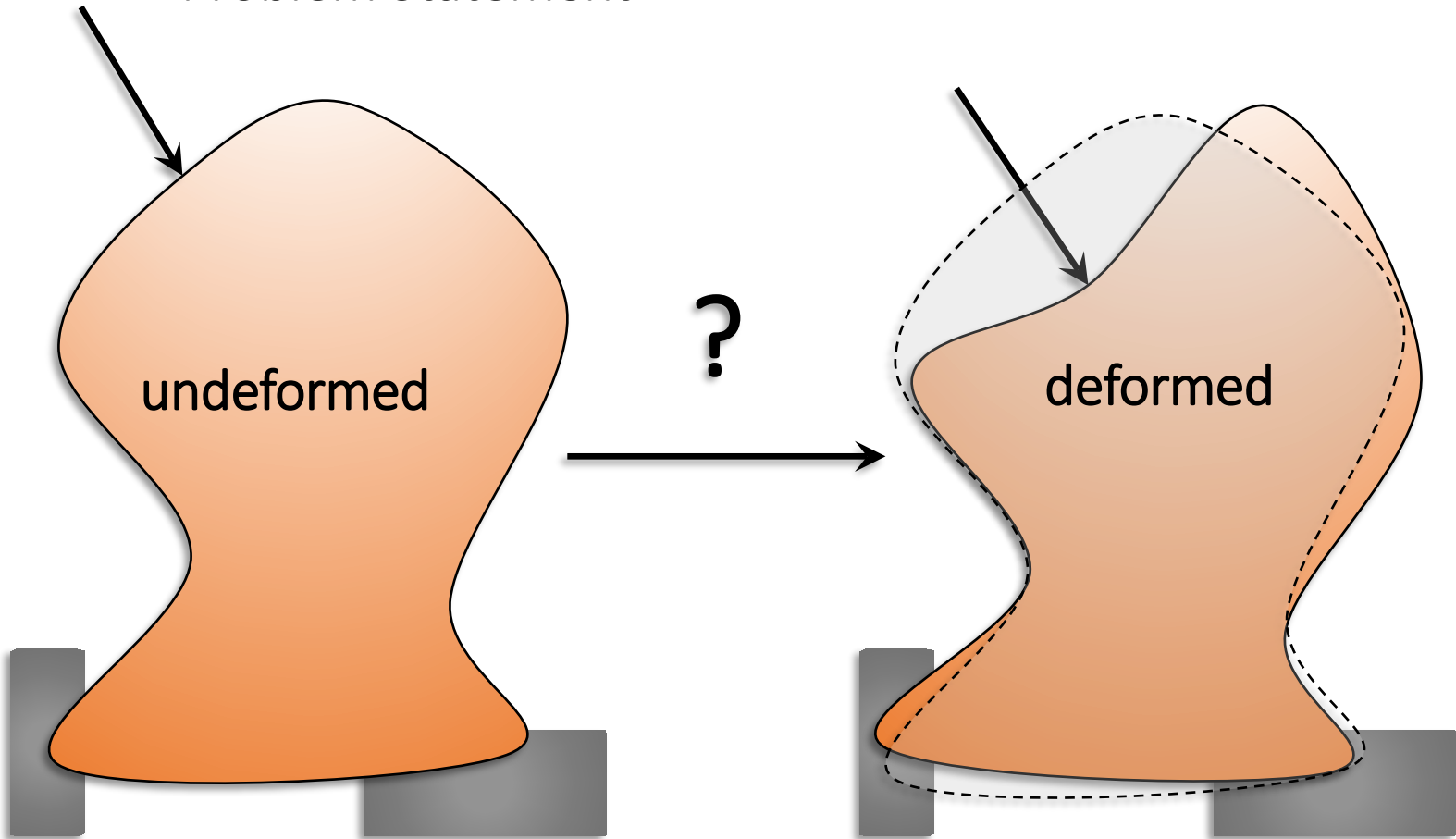
Main Features

- *Acting Loads*
- *Boundary:* The surface enclosing the geometry
- *Solid:* Interior + Boundary
- *Boundary conditions:* prescribed displacements/tractions on the boundary



FE Analysis in brief...

Problem Statement



Chapter 1: FE Analysis in brief...

How to approximate complex structures/domains?

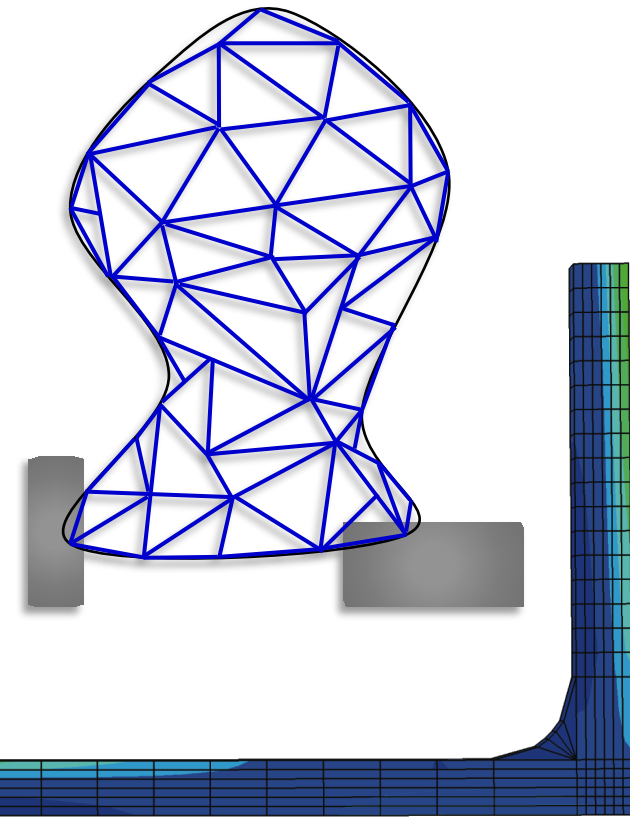
The geometry of a structure is *discretized* when it is split into a mesh of finite elements. The discretization introduces another approximation.

Two error sources can be identified:

- the *modelling error* and
- the *discretization error*

The modelling error is reduced by improving the conceptual and structural models, which describe the actual behaviour of the structure.

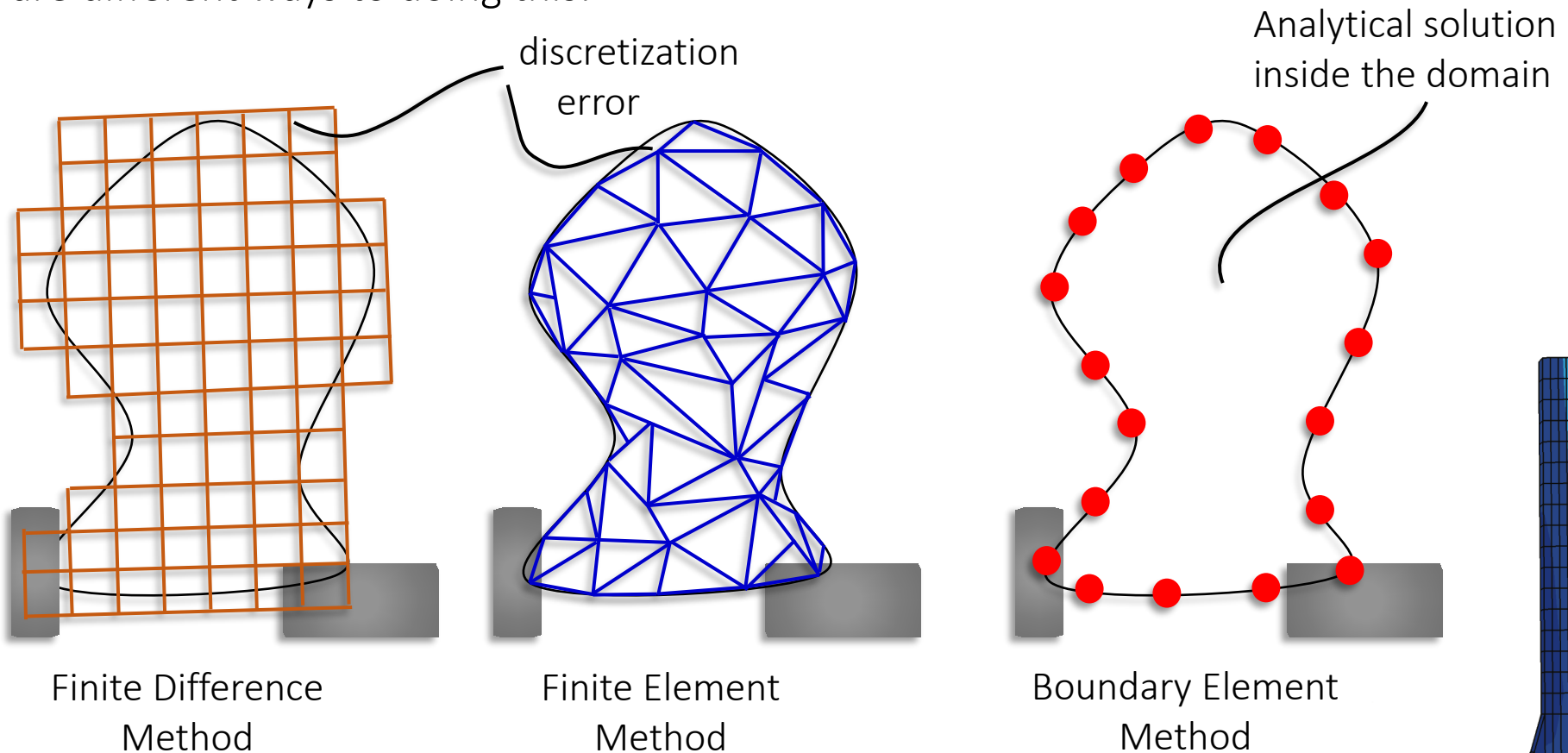
The discretization error can be reduced by using a finer mesh (i.e. more elements), or else by increasing the accuracy of the finite elements chosen (using for instance higher order polynomial expansions for approximating the displacement field within each element).



Chapter 1: FE Analysis in brief...

How to approximate complex structures/domains?

Break down the system to smaller components by *discretizing* the structure. Here are different ways to doing this:



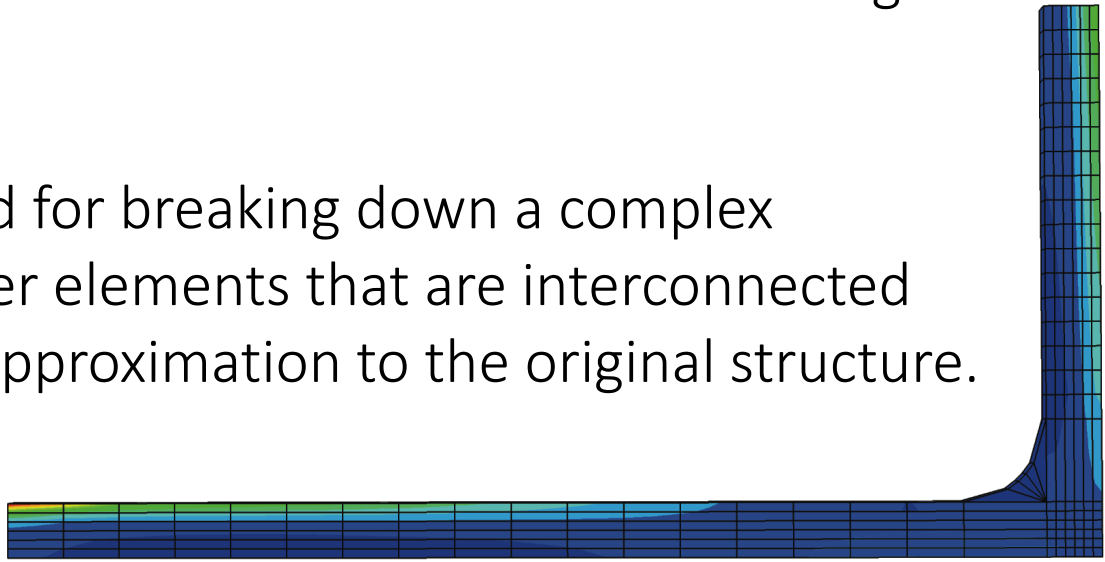
FE Analysis in brief...

How can we use FEM to solve this problem?

The Finite Element Method (FEM) is a numerical method for solution of systems where the governing equations are expressed in the form of partial differential equations (PDEs).

It essentially degenerates the original PDE problem to a system of algebraic equations, which are solvable via traditional linear algebra techniques.

Simply put, FEM is a method for breaking down a complex problem/domain into smaller elements that are interconnected and assembled to form an approximation to the original structure.

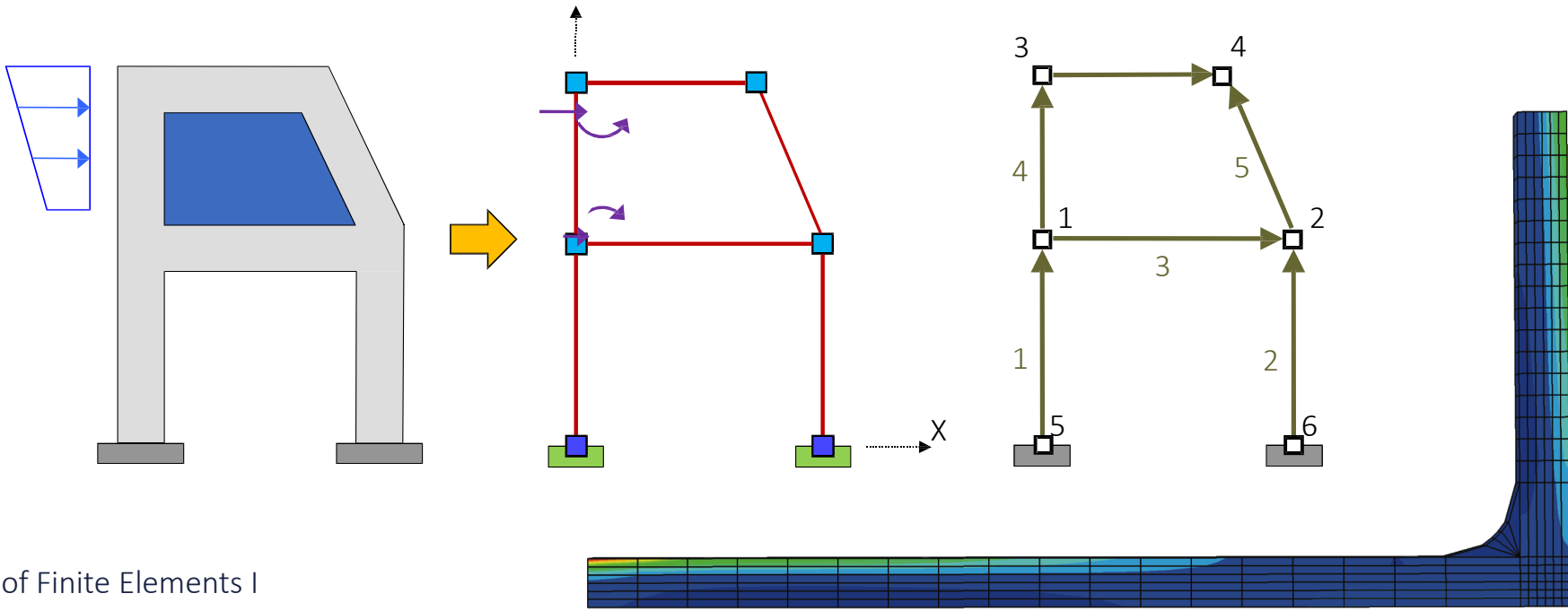


FE Analysis in brief...

Simply put, FEM is a method for breaking down a complex problem/domain into smaller elements that are interconnected and assembled to form an approximation to the original structure.

Does this ring a bell?

Baustatik II – The Direct Stiffness Method



MFE Historical Evolution

How was FEM historically formed?

(source: [Carlos A. Felippa](#))

The MFE is the confluence of three ingredients: **matrix structural analysis**, the **variational approach** and a **computer**.

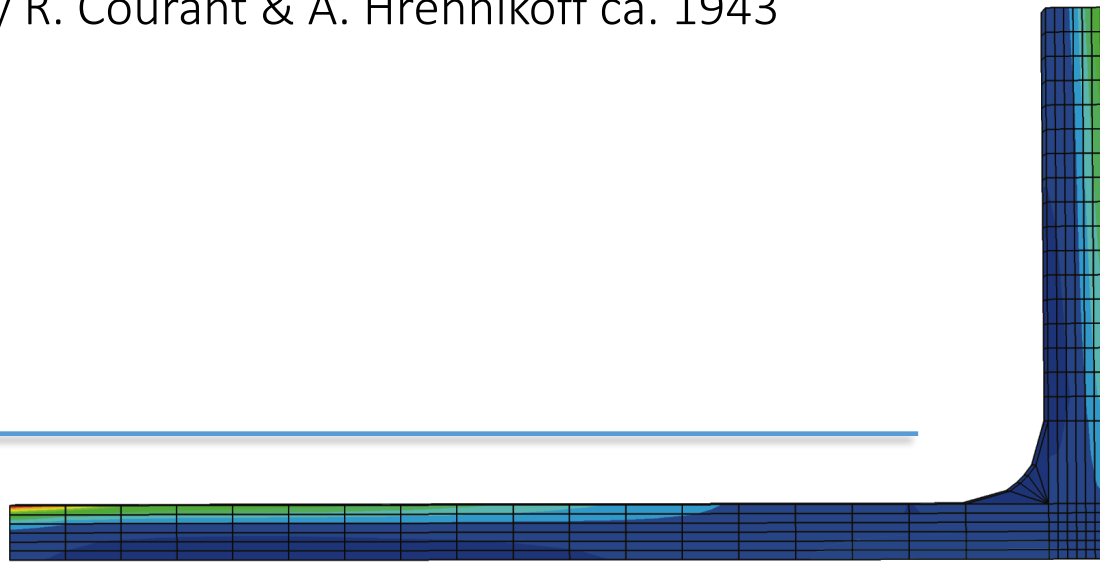
Theoretical Formulation

1. “Lösung von Variationsproblemen” by W. Ritz in 1908
2. “Weak formulation” by B. Galerkin in 1915
3. “Mathematical foundation” by R. Courant & A. Hrennikoff ca. 1943

Discretization of
the continuum



1943



MFE Historical Evolution

How was FEM historically formed?

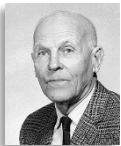
(source: [Carlos A. Felippa](#))

1950s, M.J. Turner at Boeing came up with the Direct Stiffness Method (DSM)

“Turner got Boeing to commit resources to it while other aerospace companies (Douglas, Lockheed, Rockwell, ...) were mired in the Force Method swamp. He oversaw the development of the first continuum based finite elements, the consequent creation and expansion of the DSM, and its first application to nonlinear problems”

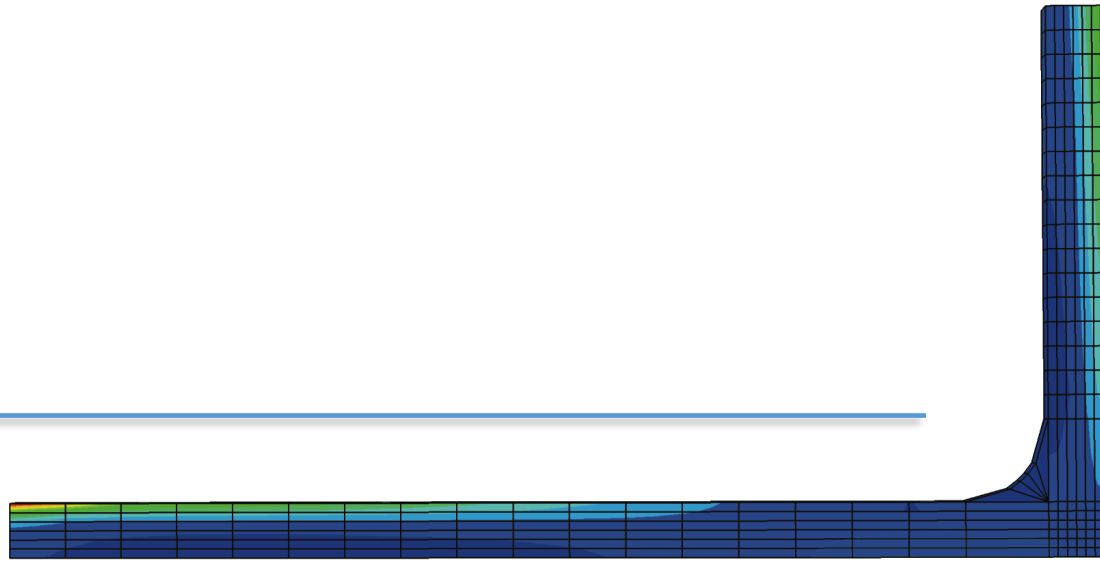
continuum discretization

DSM emergence



1943

1950



MFE Historical Evolution

Who were the “populizers”?

(source: [Carlos A. Felippa](#))

1. Matrix formulation of structural analysis by **Agyris** in **1954**

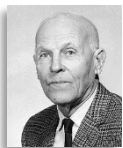
“As a consultant to Boeing in the early 1950s, Argyris, a Force Method expert then at Imperial College, he interacted with Turner & was the first to construct a displacement-assumed continuum element.”

2. The term “Finite Element” was coined by **Clough** in **1960**

“R.W. Clough and H.C. Martin, then junior professors at U.C. Berkeley and U. Washington, respectively, spent “faculty internship” summers at Turner’s group and collaborated on a celebrated paper, which is broadly considered the start of the present FEM.”

3. First book on FEM by **Zienkiewicz, Taylor, and Zhu** 1967

continuum discretization



1943

DSM emergence



1950

matrix formulation



1954

FEM coined



1956

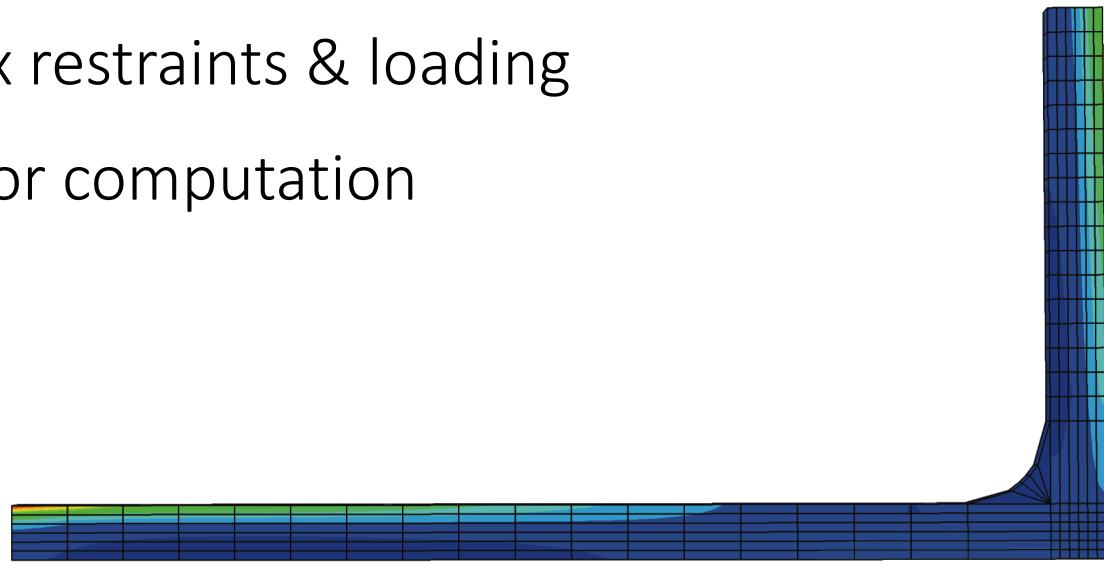
[FEM Book](#)



1967

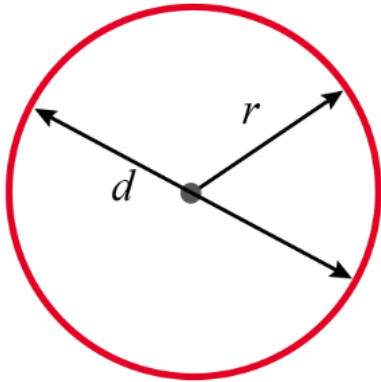
FEM is a big success story, because it...

1. can handle complex geometries
2. can handle a wide variety of engineering problems
 - mechanics of solids & fluids
 - dynamics/heat/electrostatic problems...
3. can handle complex restraints & loading
4. is very well suited for computation



What is a Finite Element?

(a)



Find the perimeter L of a circle of diameter d

We know that $L = \pi d$.

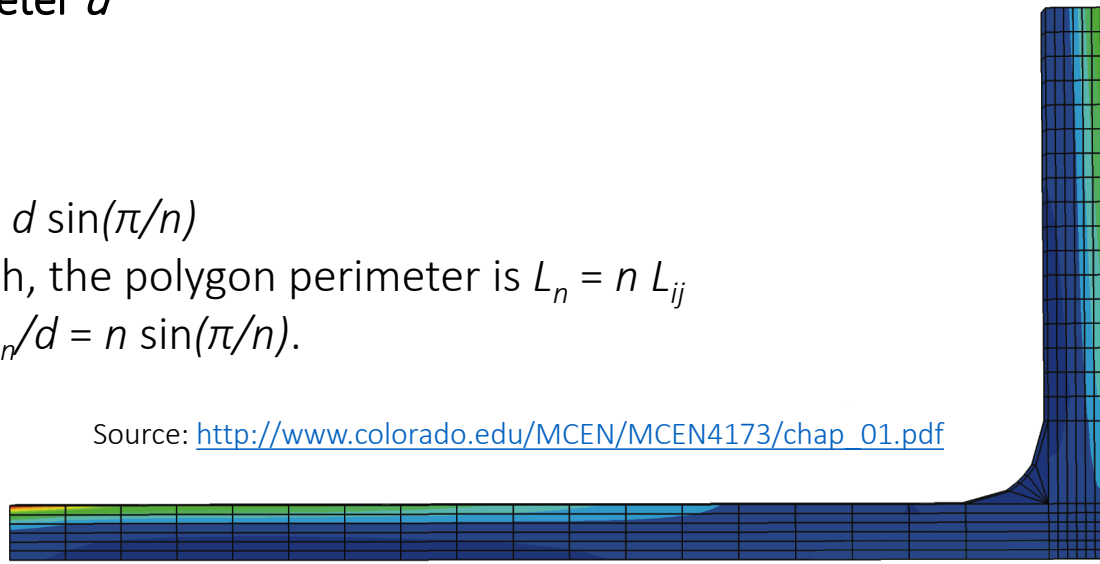
Approximation via Inscribed Polygons

The element length is $L_{ij} = 2r \sin(\pi/n) = d \sin(\pi/n)$

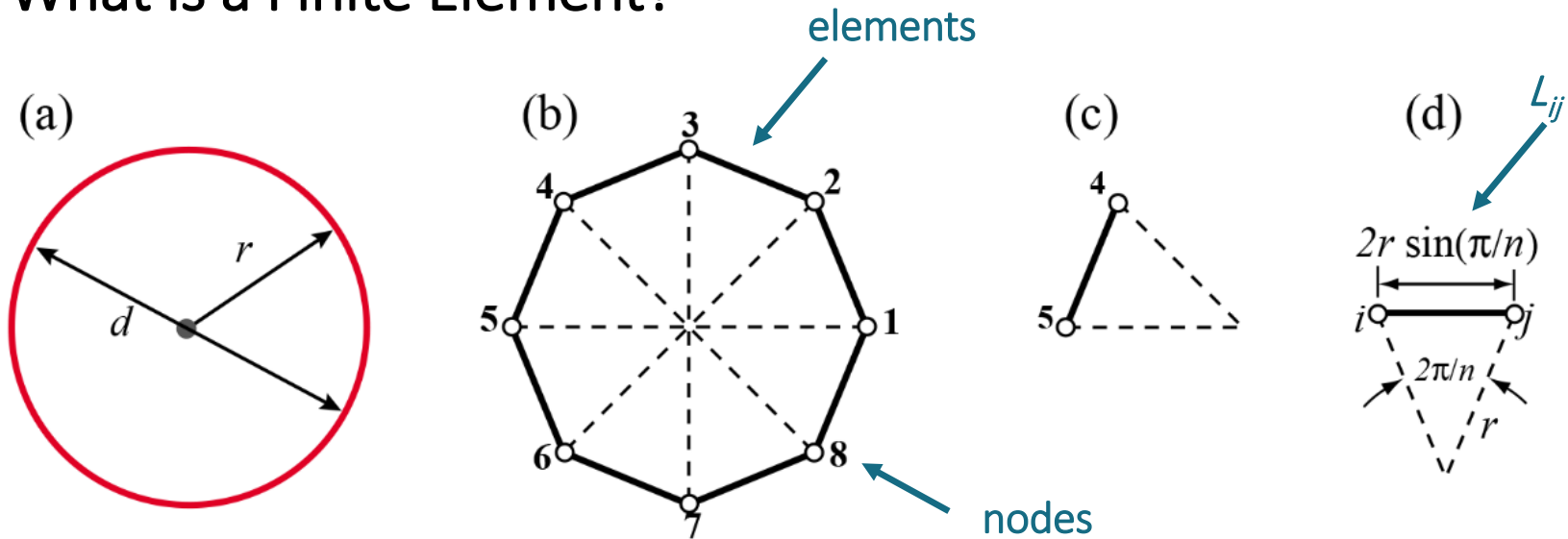
Since all elements have the same length, the polygon perimeter is $L_n = n L_{ij}$

Hence the approximation to π is $\pi n = L_n/d = n \sin(\pi/n)$.

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_01.pdf



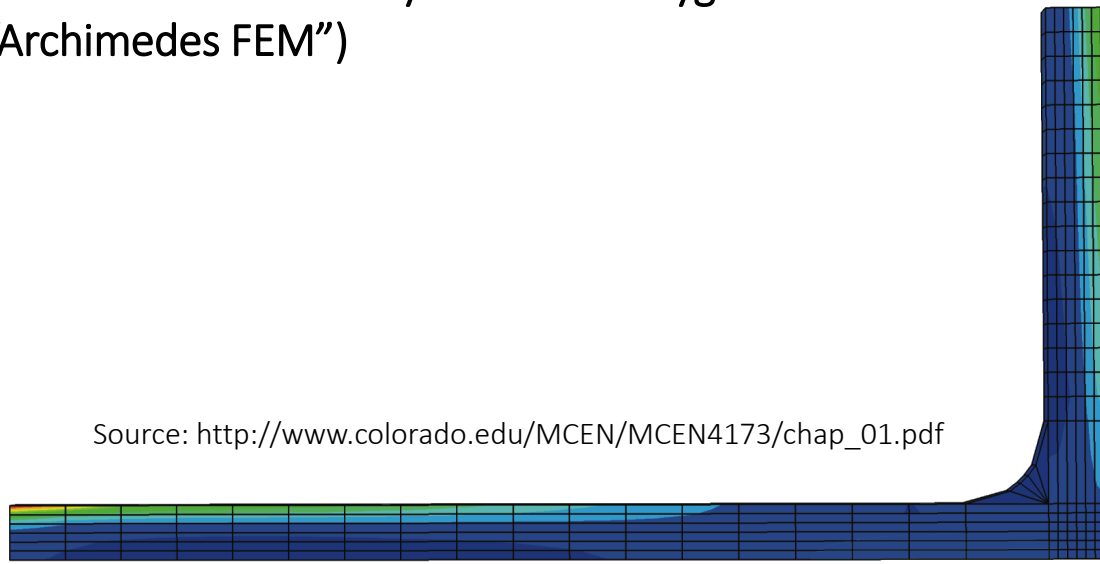
What is a Finite Element?



n	$\pi_n = n \sin(\pi/n)$
1	0.0000000000000000
2	2.0000000000000000
4	2.828427124746190
8	3.061467458920718
16	3.121445152258052
32	3.136548490545939
64	3.140331156954753
128	3.141277250932773
256	3.141513801144301

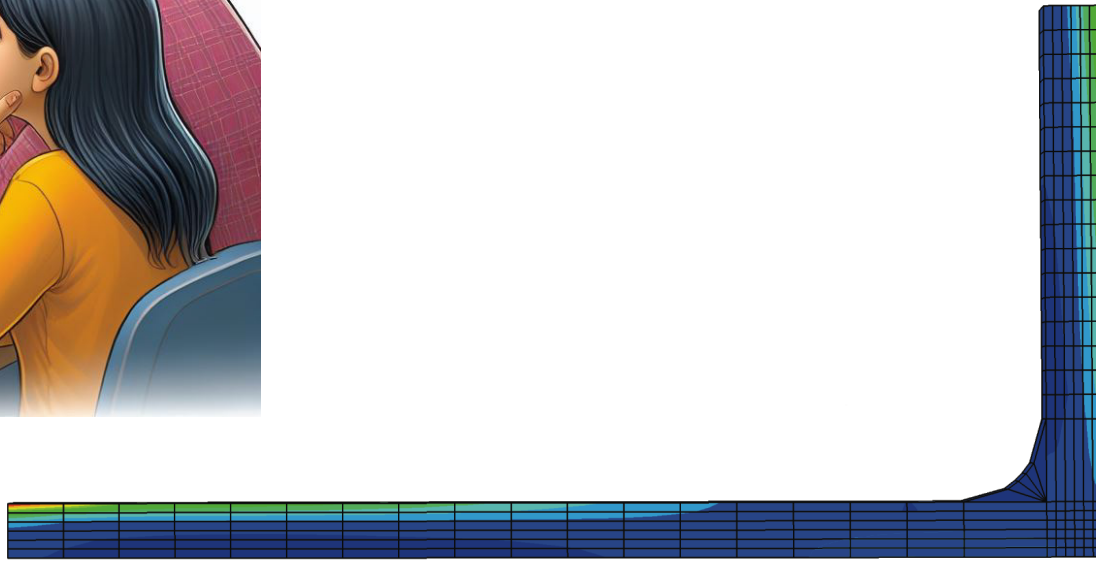
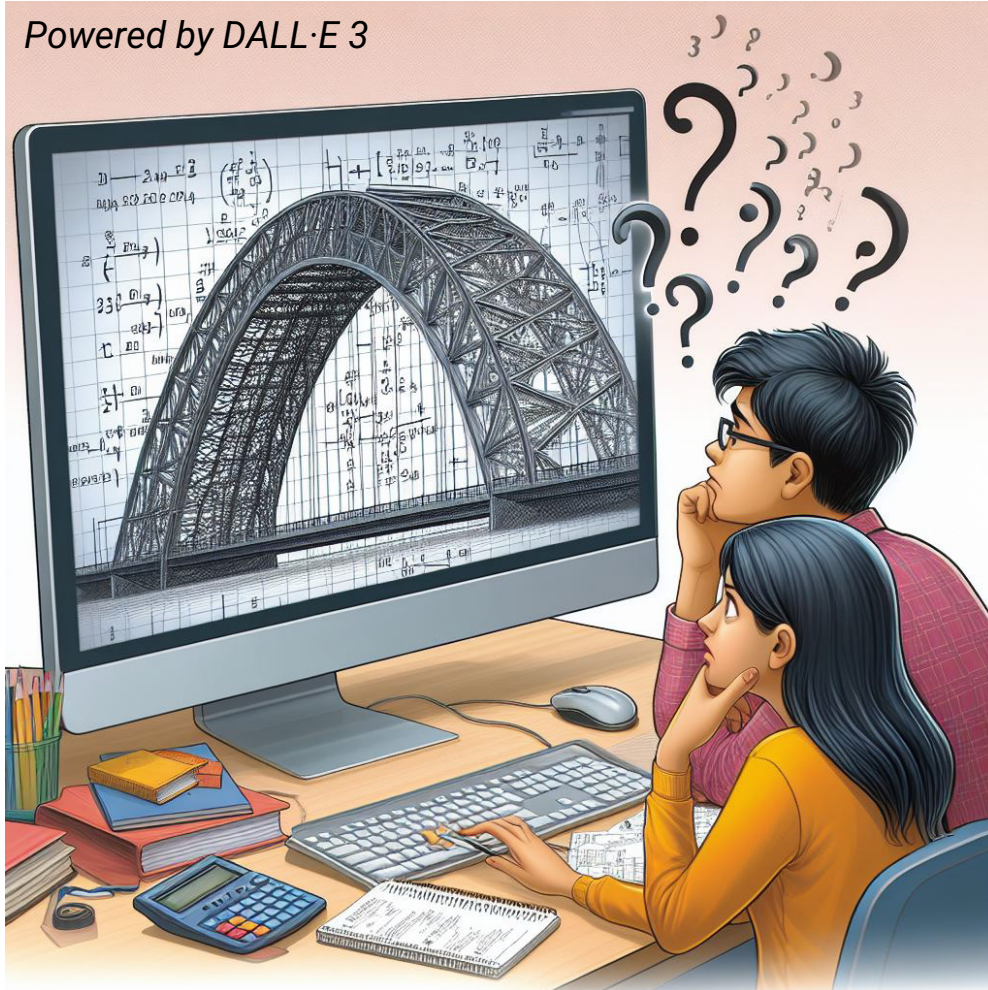
Rectification of Circle by Inscribed Polygons
("Archimedes FEM")

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_01.pdf



Why & When to use FEM?

Powered by DALL·E 3



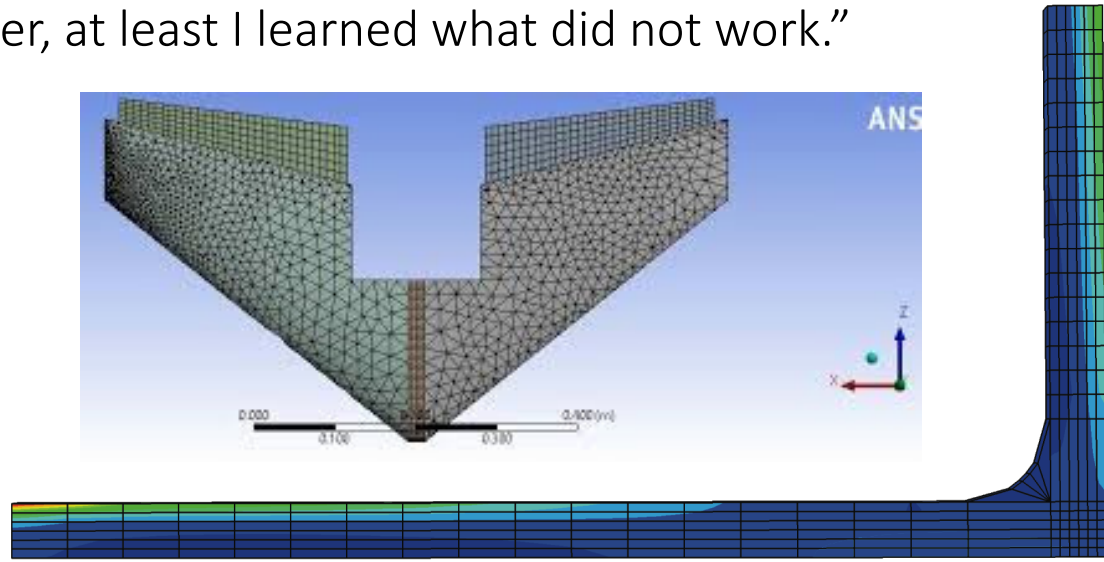
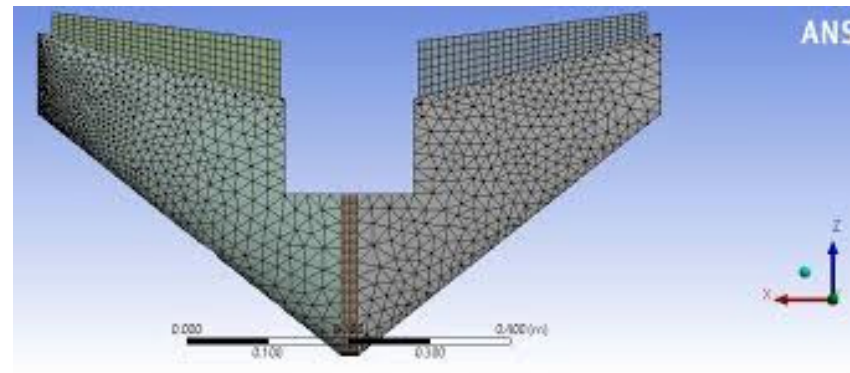
Why & When to use FEM?

Let's follow Clough's trip:

(source: [Carlos A. Felippa](#))

“When I arrived in the summer of 1952, Jon Turner asked me to work on the **vibration analysis of a delta wing structure**. Because of its triangular plane form, **this problem could not be solved by procedures based on standard beam theory**; so, I spent the summer of 1952 trying to formulate a delta wing model built up of an assembly of one-dimensional beams and struts.

However, the results of deflection analysis based on this type of mathematical model were in very poor agreement with data obtained from laboratory tests of a scale model of a delta wing. My final conclusion was that my summer's work was a total failure — however, at least I learned what did not work.”



Why & When to use FEM?

Let's follow Clough's trip:

(source: [Carlos A. Felippa](#))

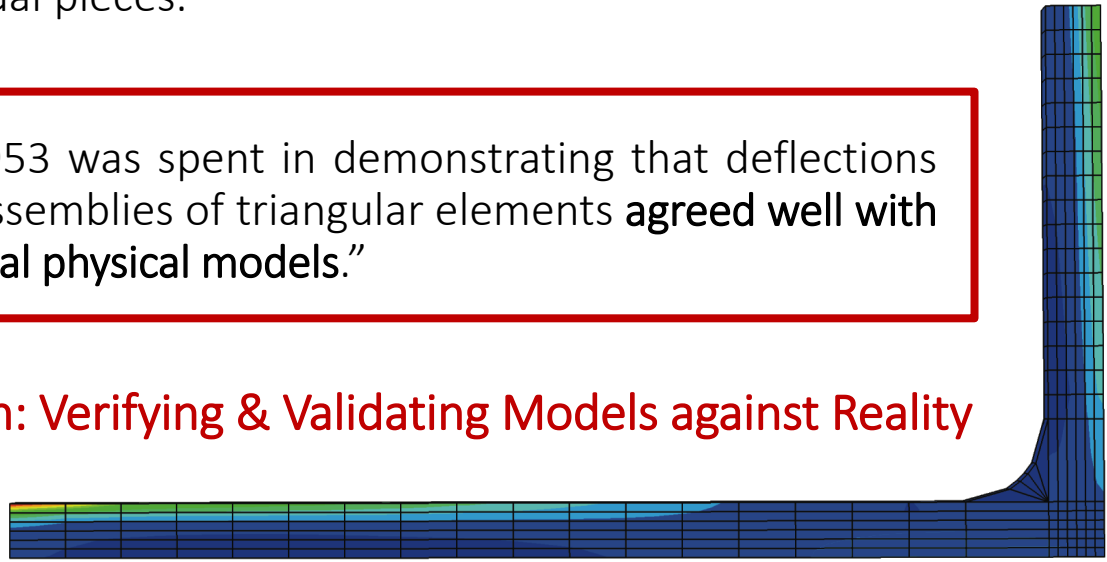
“The most important development during the winter was that Jon suggested we try to **formulate the stiffness properties of the wing by assembling plane stress plates of either triangular or rectangular shapes**

... the triangular form was much more useful because such plates could be assembled to approximate structures of any configuration.

Then the stiffness of the complete structure was obtained by appropriate addition from the contributions of the individual pieces.

The remainder of the summer of 1953 was spent in demonstrating that deflections calculated for structures formed of assemblies of triangular elements **agreed well with laboratory measurements on the actual physical models.**”

Motivation: Verifying & Validating Models against Reality

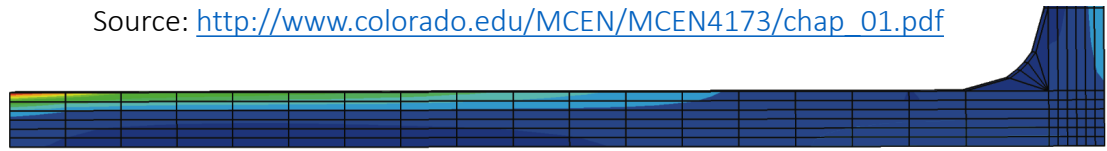


How does it work?

Physical Model

Describe the problem: Simplify a real engineering problem into a problem that can be solved by FEA

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_01.pdf



How does it work?

Physical Model

Describe the problem: Simplify a real engineering problem into a problem that can be solved by FEA

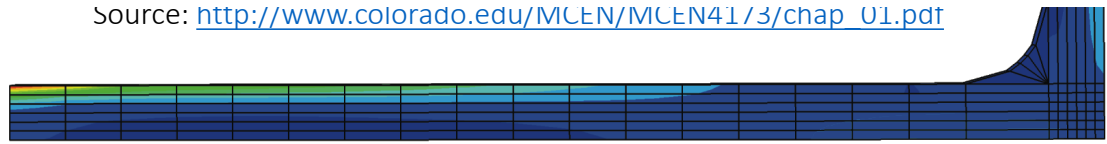


Pre-processor

FE model

Discretize/mesh the solid, define material properties, apply boundary conditions

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_U1.pdf



How does it work?

Physical Model

Describe the problem: Simplify a real engineering problem into a problem that can be solved by FEA



Pre-processor

FE model

Discretize/mesh the solid, define material properties, apply boundary conditions

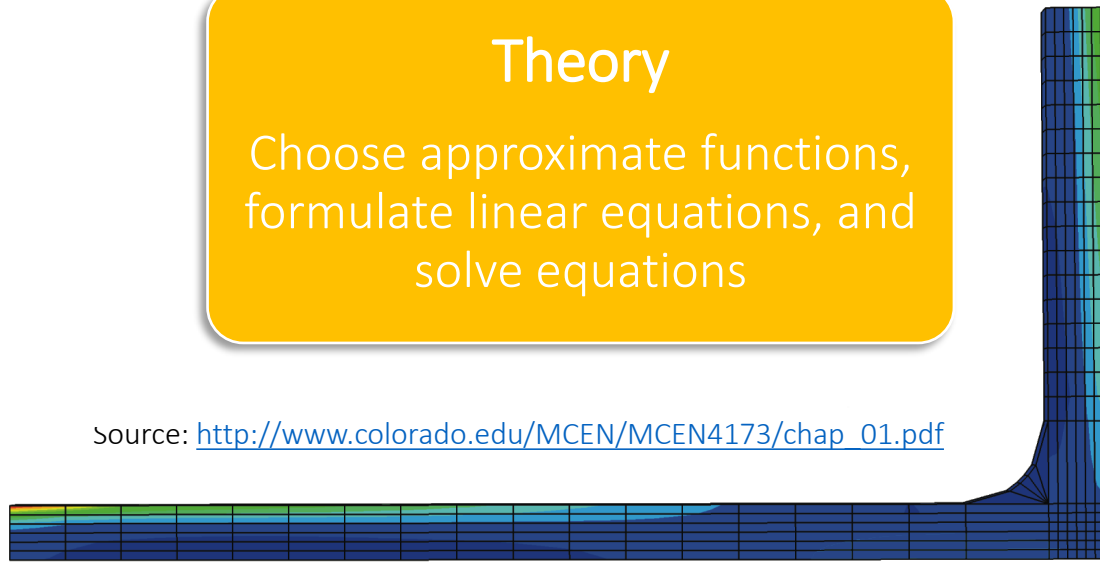


Solver

Theory

Choose approximate functions, formulate linear equations, and solve equations

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_01.pdf



How does it work?

Physical Model

Describe the problem: Simplify a real engineering problem into a problem that can be solved by FEA



Pre-processor

FE model

Discretize/mesh the solid, define material properties, apply boundary conditions



Often viewed as black box

Post-processor

Results

Obtain, visualize and explain the results

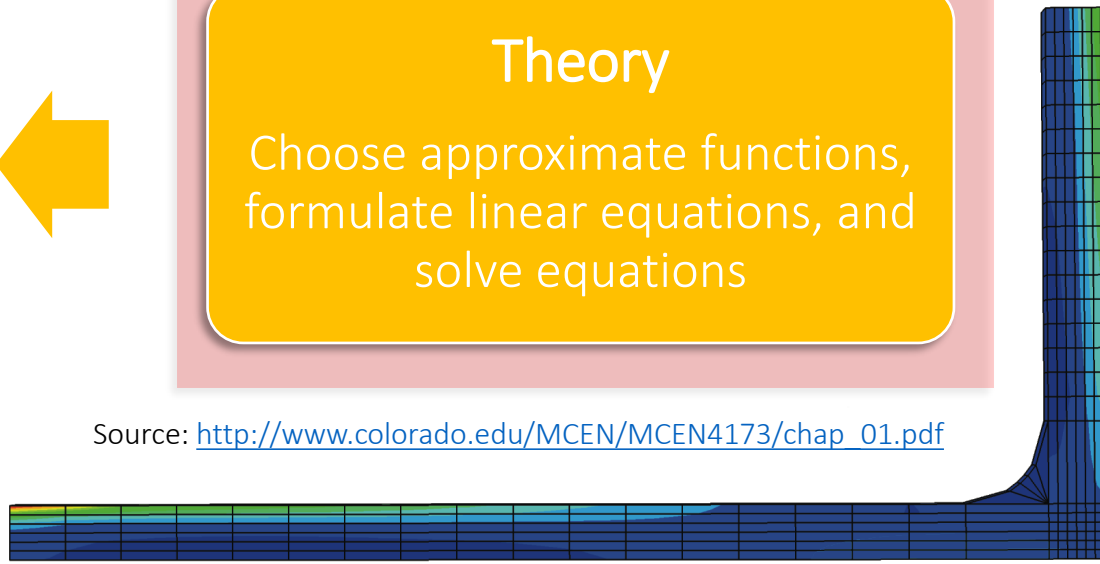


Solver

Theory

Choose approximate functions, formulate linear equations, and solve equations

Source: http://www.colorado.edu/MCEN/MCEN4173/chap_01.pdf



How does it work?

FE Workflow

- Start from a complex problem
- Break it into pieces (elements)
- Use simple math to represent the force displacement relationship on each piece $\mathbf{f}_i = \mathbf{k}_i \mathbf{u}_i$.

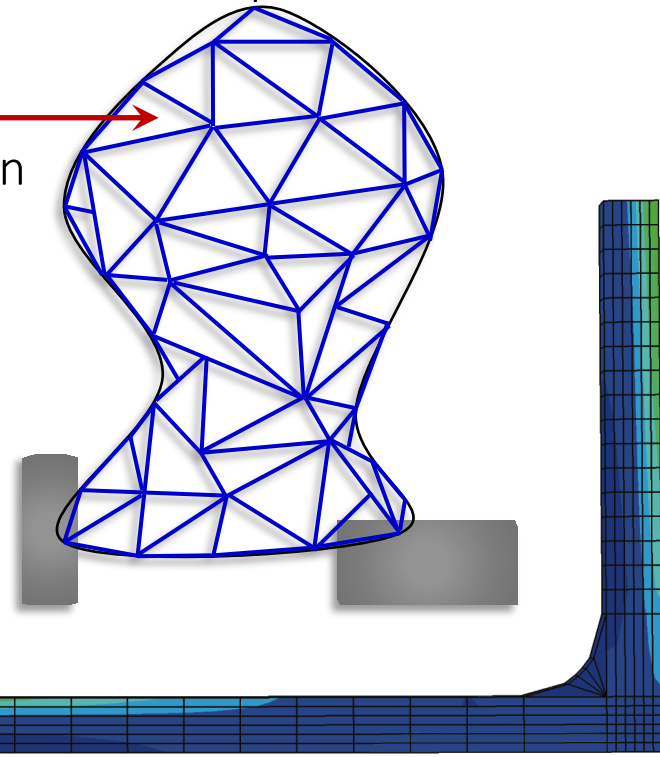
- Assemble the pieces into one global (matrix) equation

$$\mathbf{F} = \mathbf{K} \mathbf{U}$$

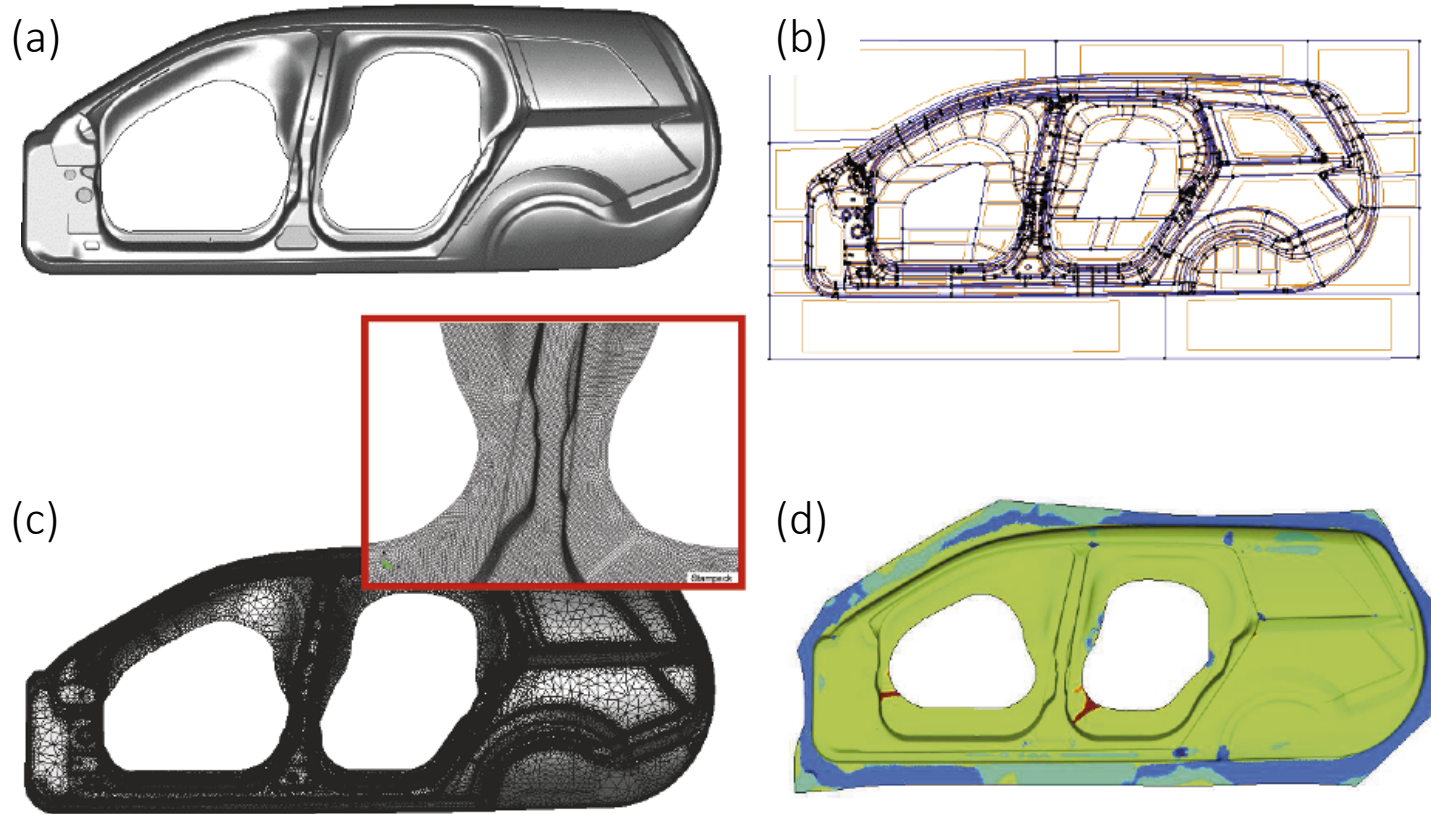
- Apply Boundary Conditions to constrain the problem

$$\mathbf{F}_{BC} = \mathbf{K}_{BC} \mathbf{U}_{BC}$$

- Solve the problem to obtain quantities of interest
e.g. displacements, stresses



The Structural Analysis Process; an example



(a) Actual geometry of an automotive panel. (b) CAD geometrical description by NURBS patches. (c) Finite element mesh of 3-noded shell triangles discretizing the panel geometry. (d) FEM numerical results of the structural analysis showing the equivalent strain distribution

**Fig 1.5 from Onate, Vol. 1 & www.quantech.es*

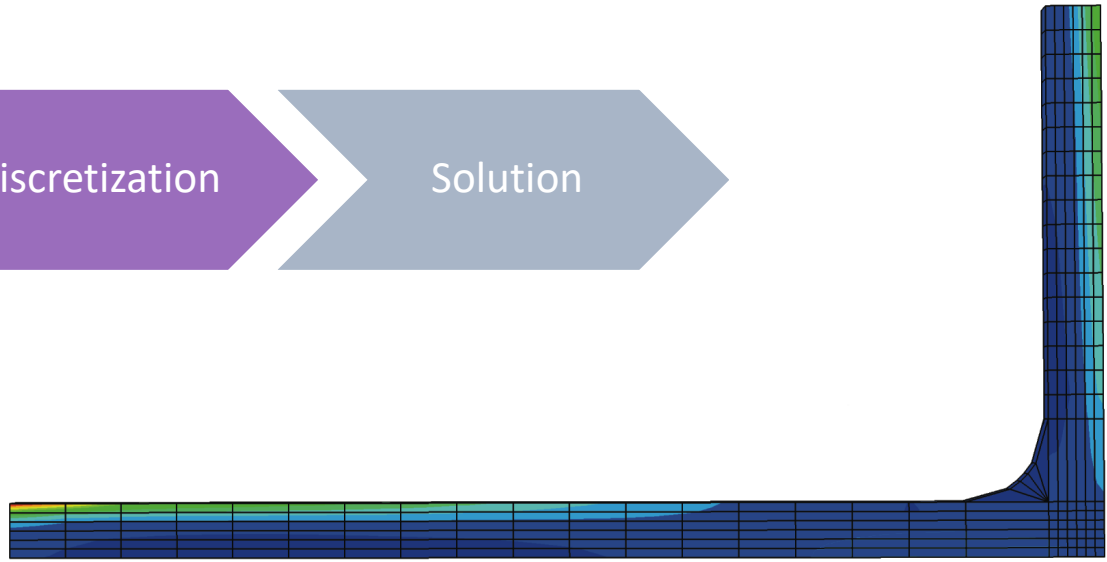
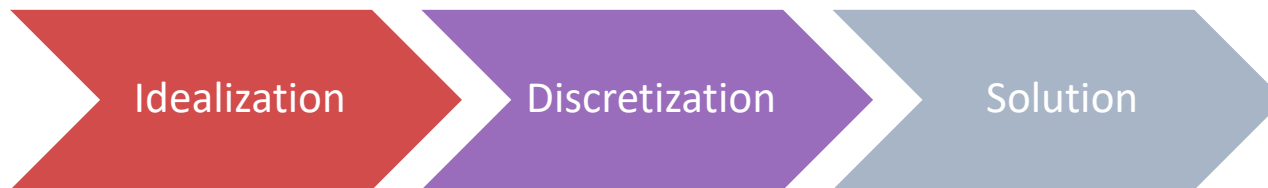
Steps of the Finite Element Method

The goal of Structural Analysis: Modeling a system by means of approximation.

How to do this?

Come up with a simplified idealization. This may oftentimes be achieved conceptually.

Basic Steps of the DSM

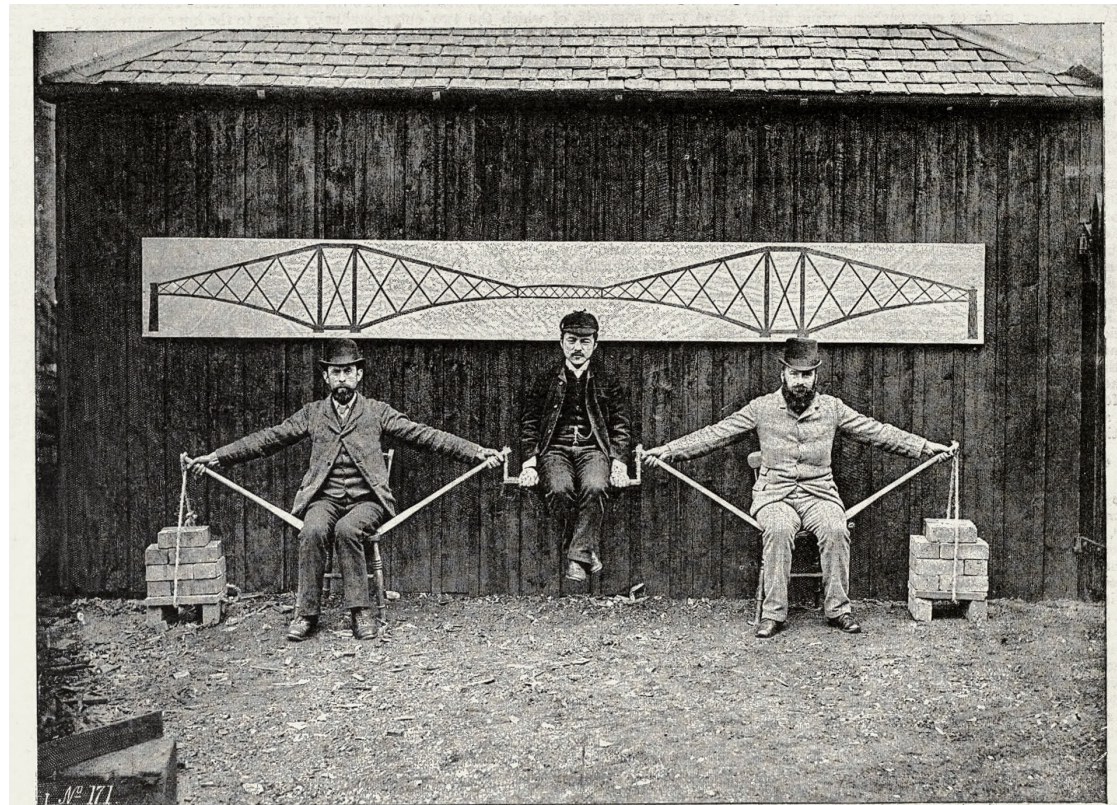


Steps of the Finite Element Method

Idealization

Revisiting our Example

How to model this ?



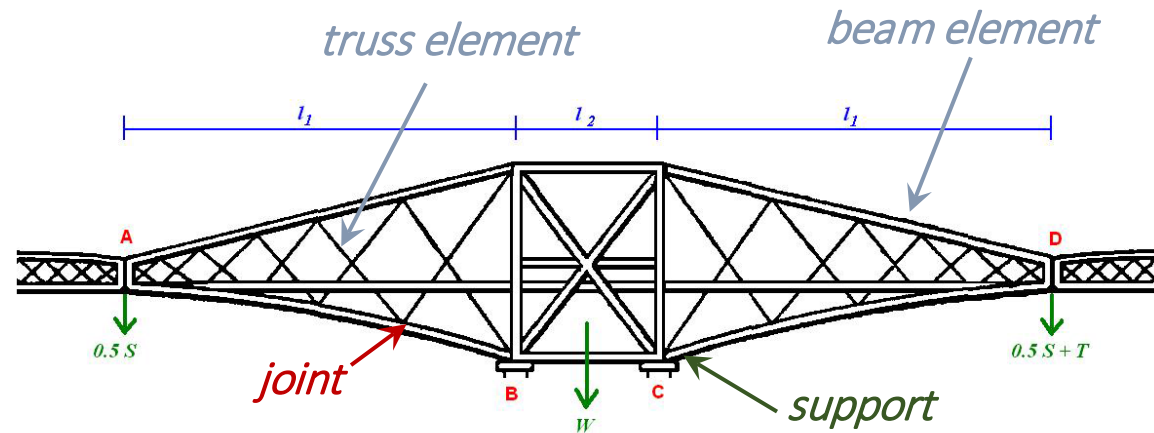
©wikimedia

Steps of the Finite Element Method



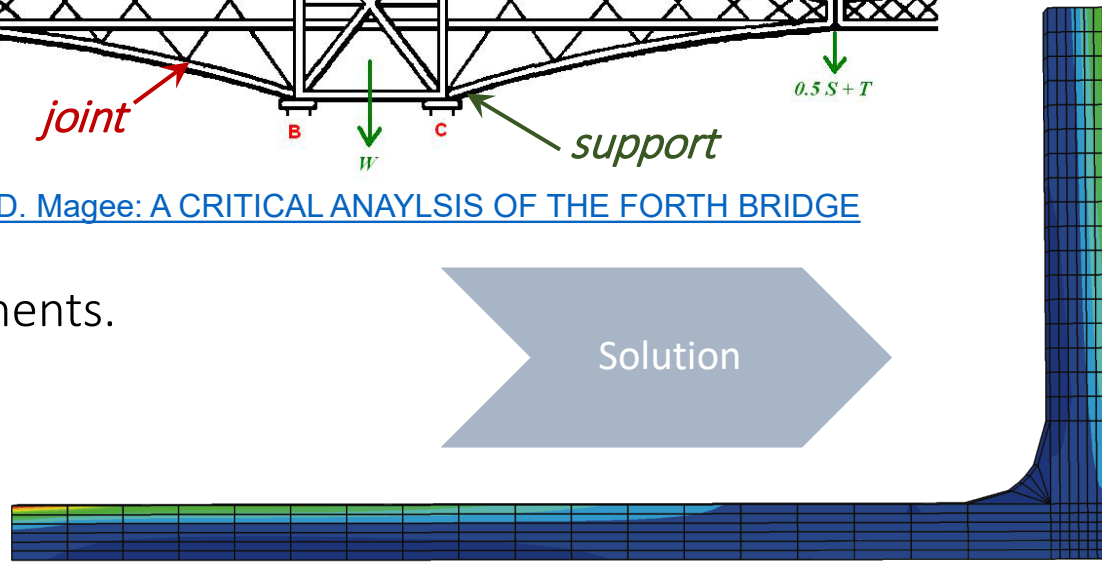
Example

How to model this ?



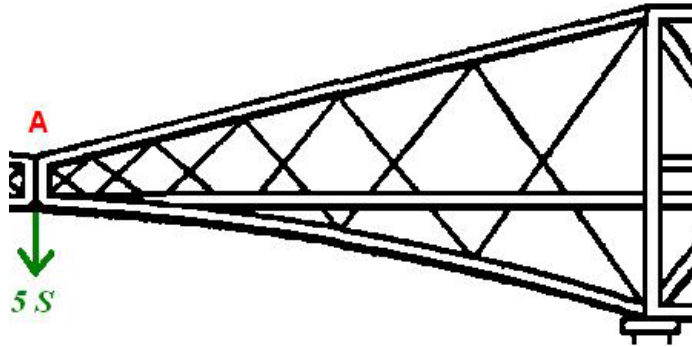
[©A.D. Magee: A CRITICAL ANALYSIS OF THE FORTH BRIDGE](#)

Break the system down to components.

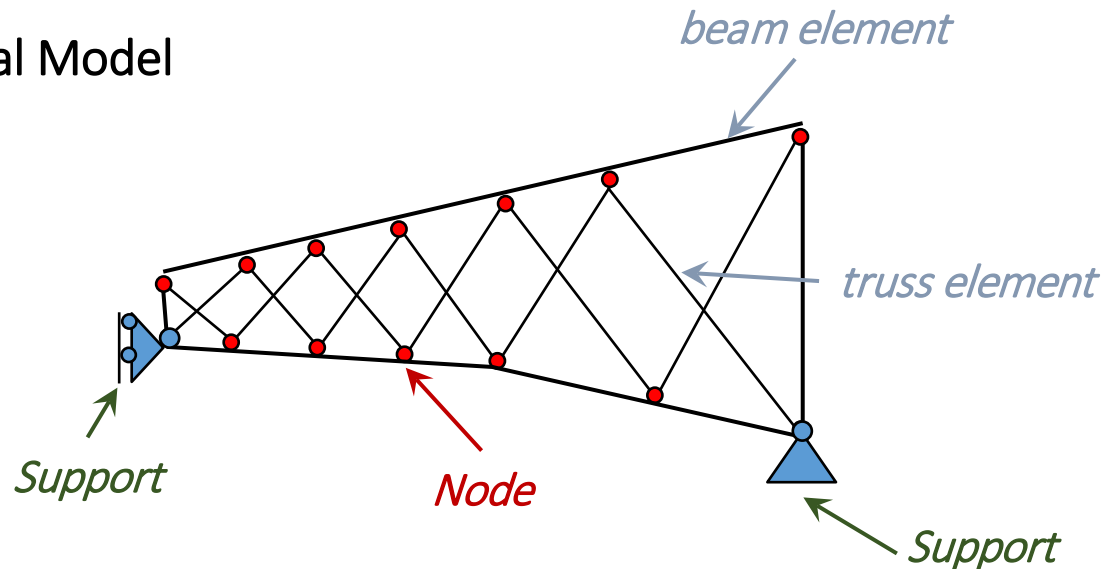


Steps of the Finite Element Method

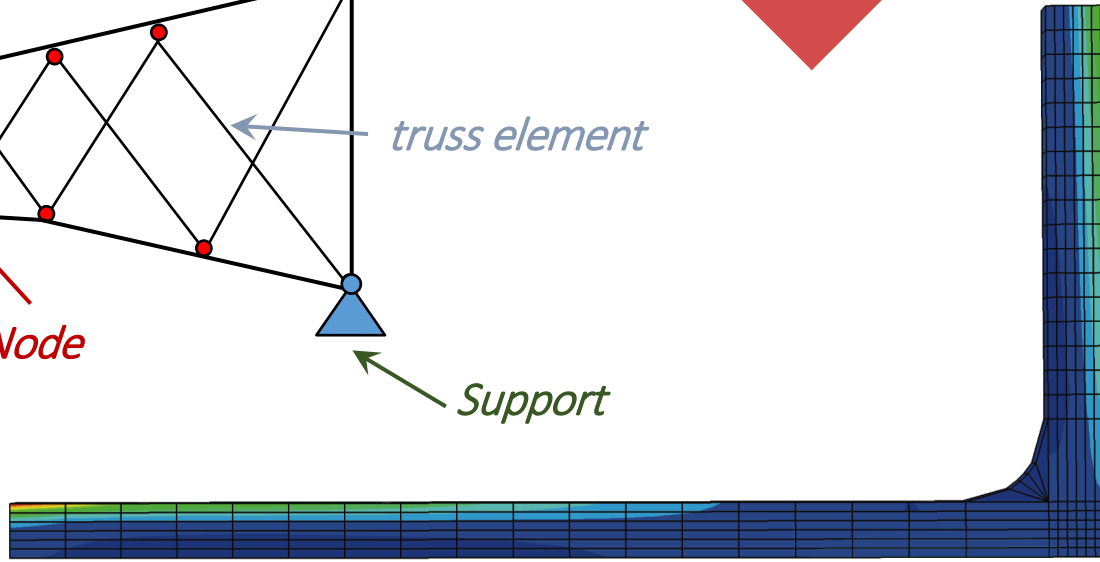
Physical System



Mathematical Model

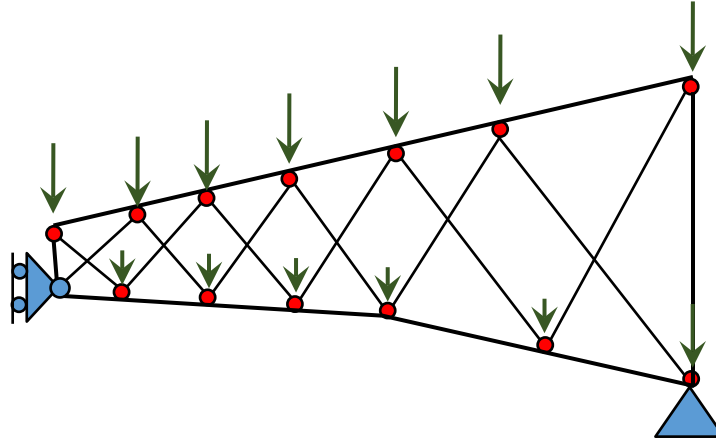


Idealization

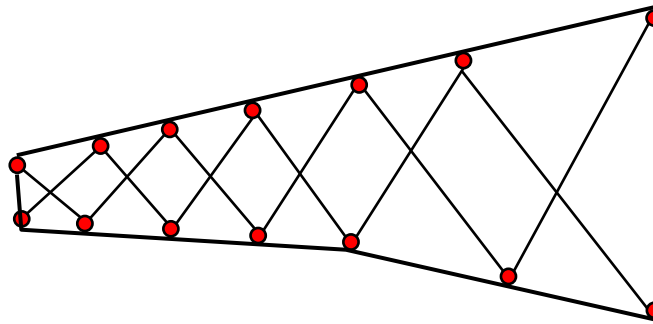


Steps of the Finite Element Method

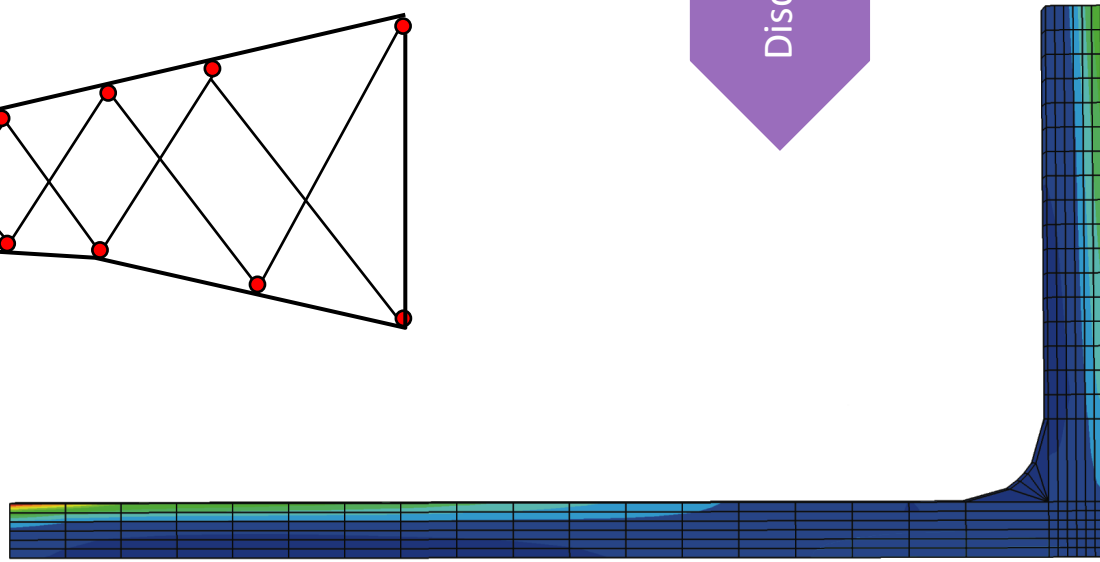
DSM Idealization



Remove loads & supports

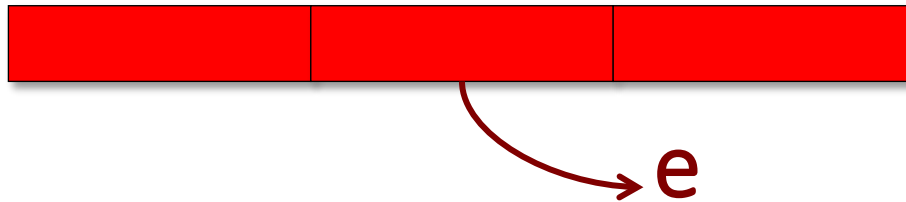


Discretization

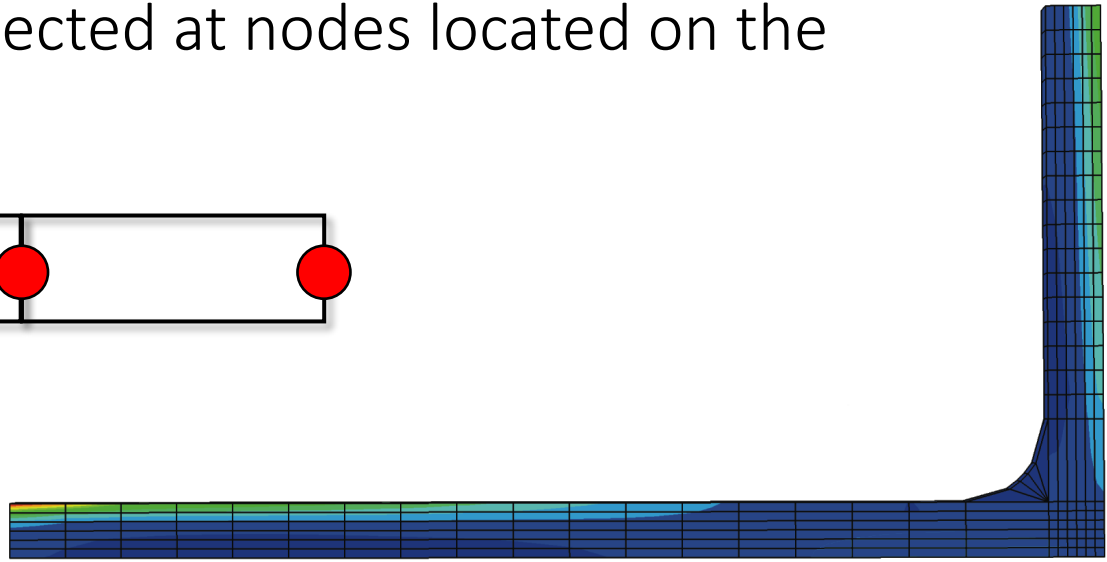
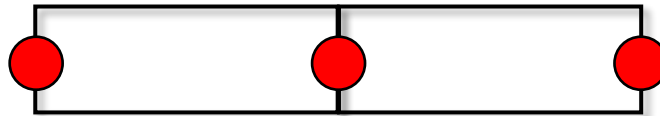


Steps of the Finite Element Method

Discretization: The continuum is discretized using a mesh of finite elements.

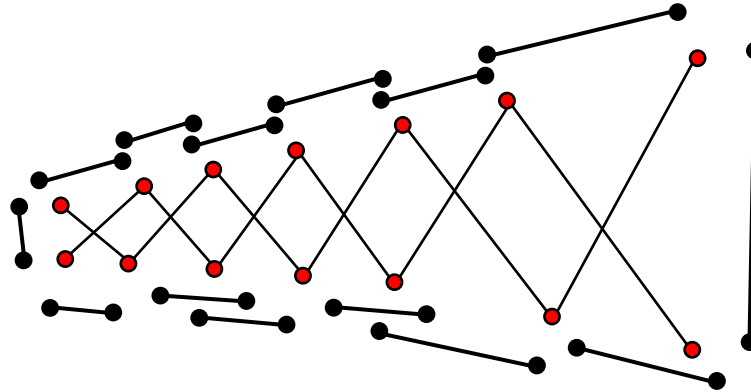


These elements are connected at nodes located on the element boundaries.



Steps of the Finite Element Method

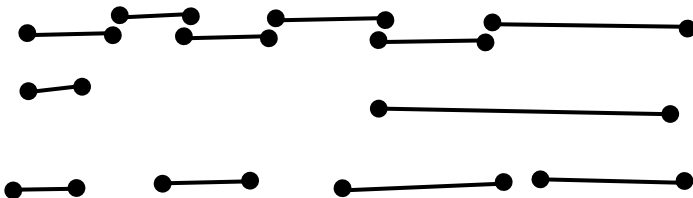
Disassemble



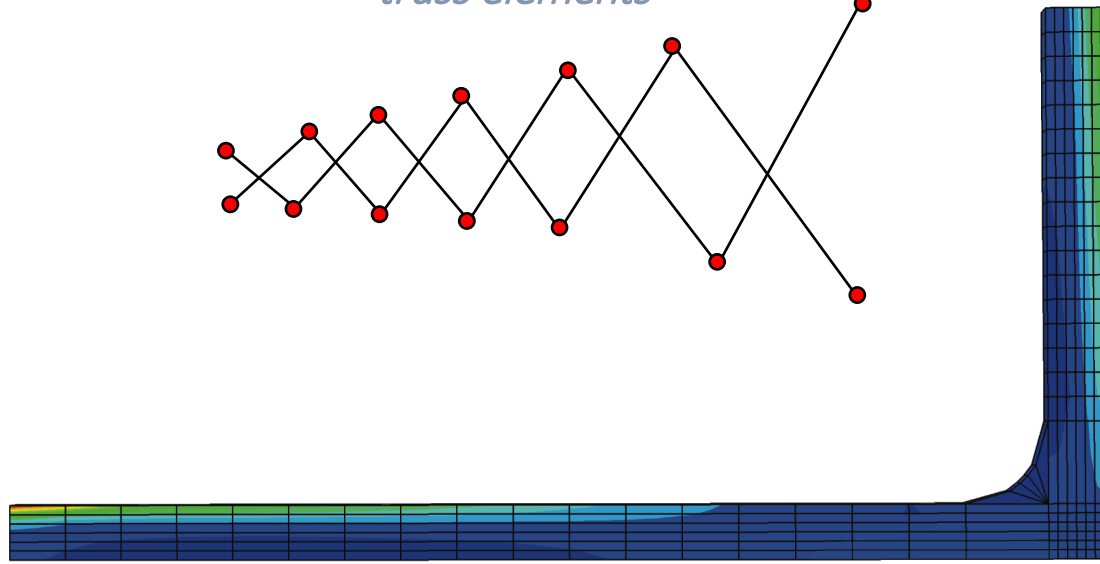
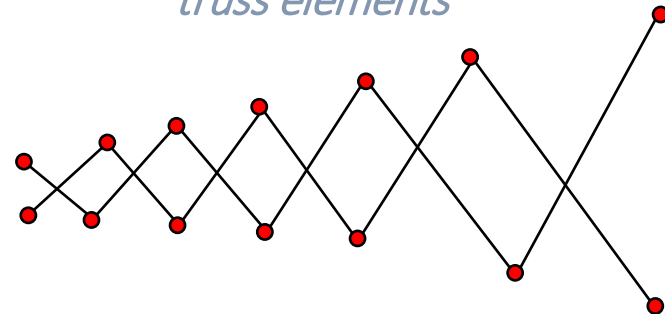
Discretization

Treat Individually

beam elements



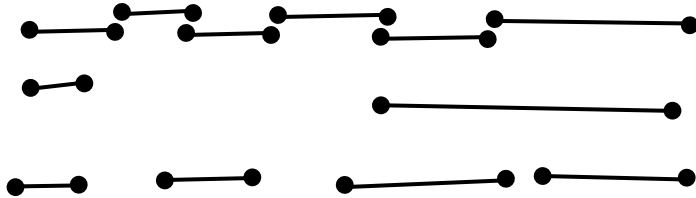
truss elements



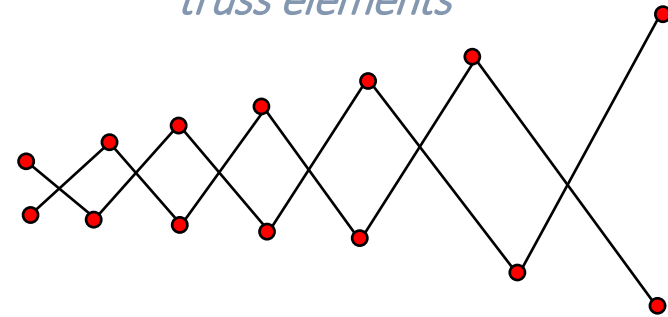
Steps of the Finite Element Method

from individual elements

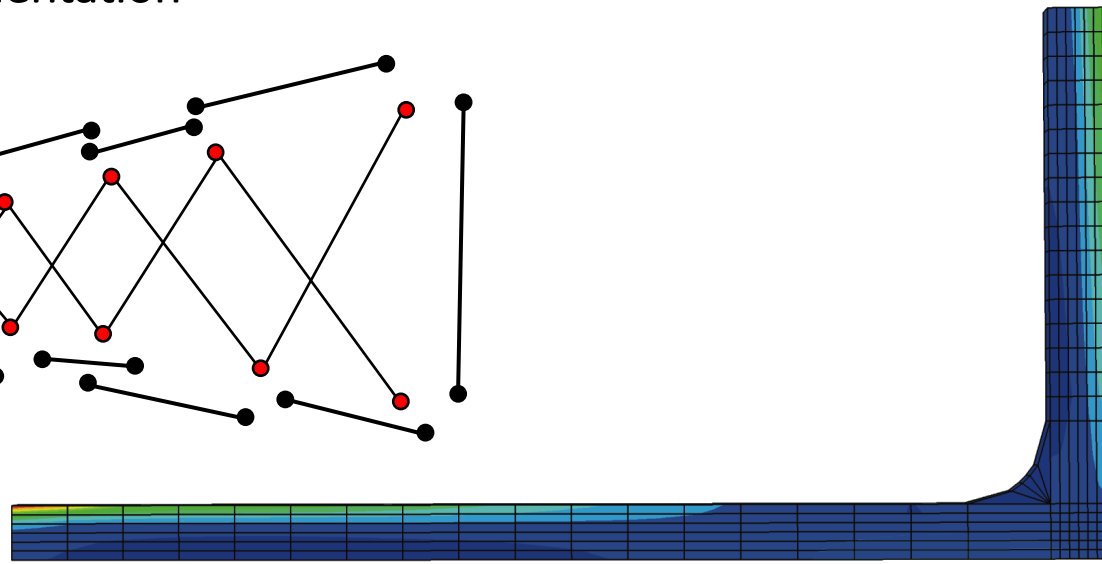
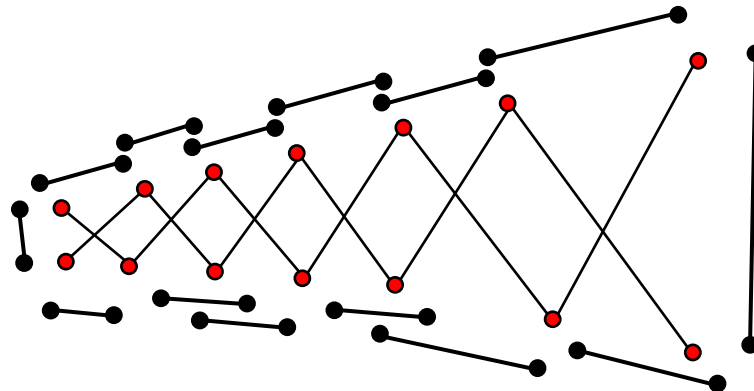
beam elements



truss elements

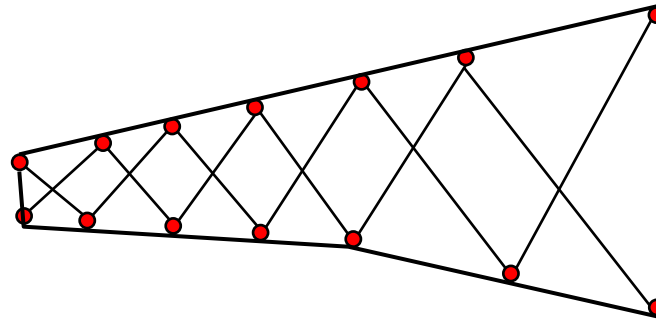


Map back to their global form/orientation

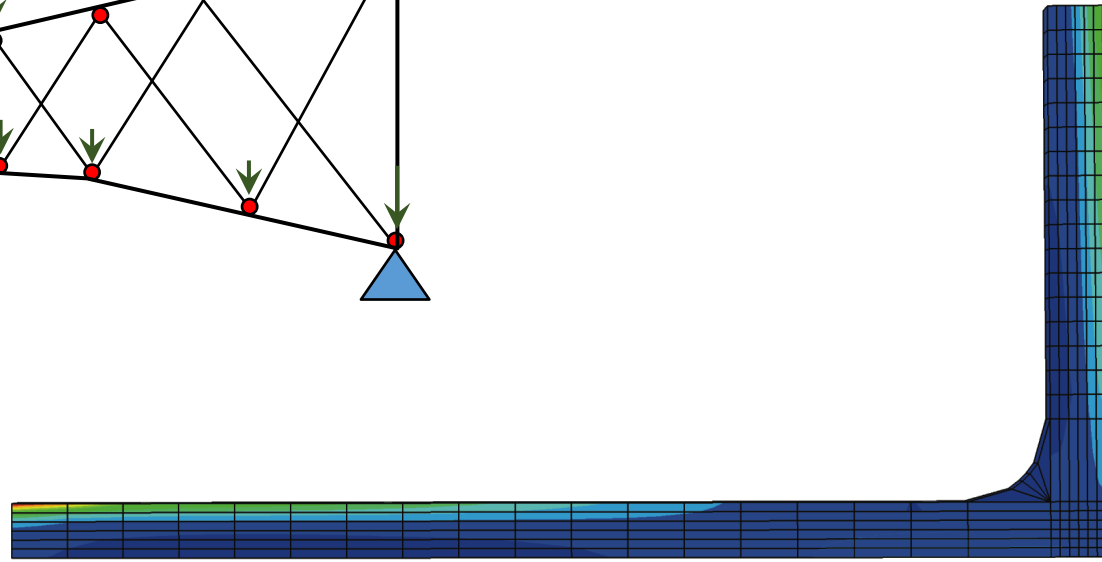
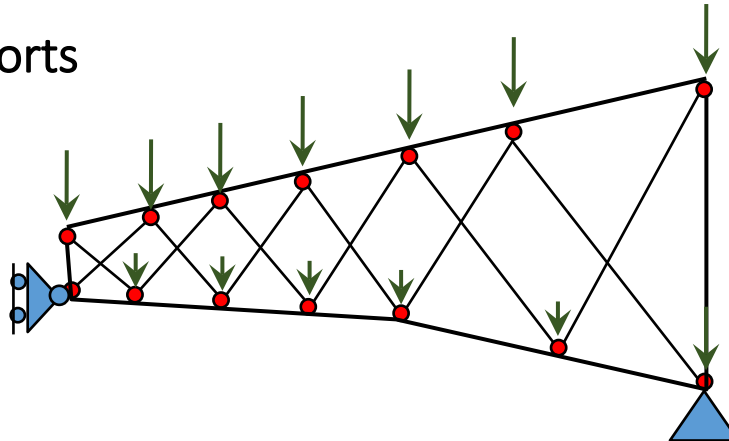


Steps of the Finite Element Method

Assemble

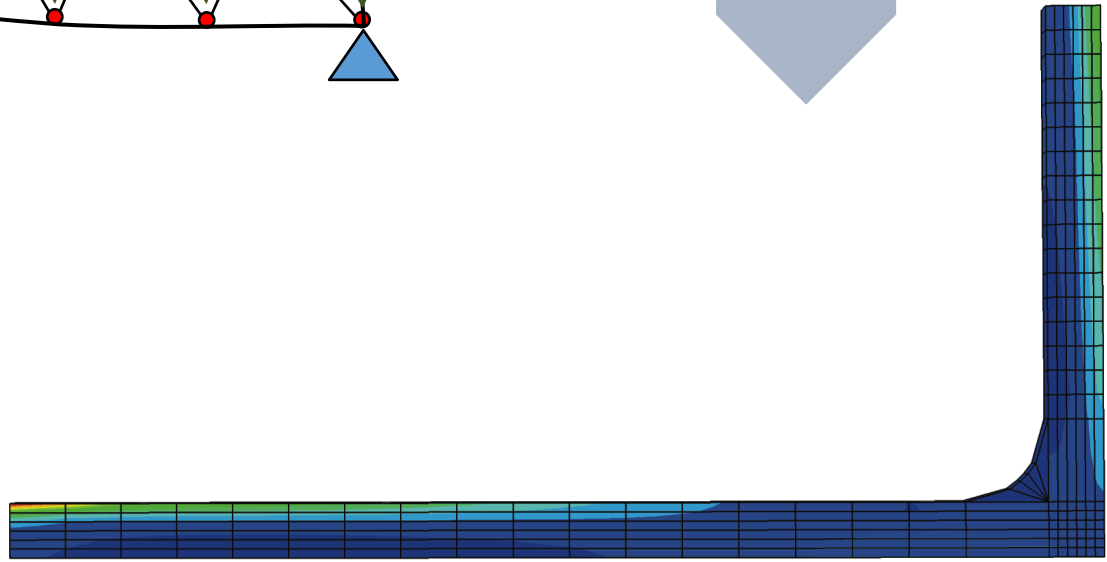
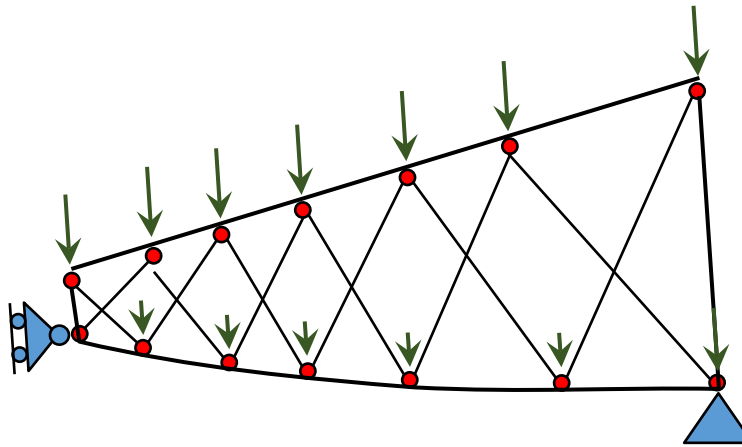


Apply loads & supports



Steps of the Finite Element Method

Solve for DOFs: displacements/rotations



Summary of the FEM Steps



Disassembly

Individualization/Localization

Member Formulation



Globalization

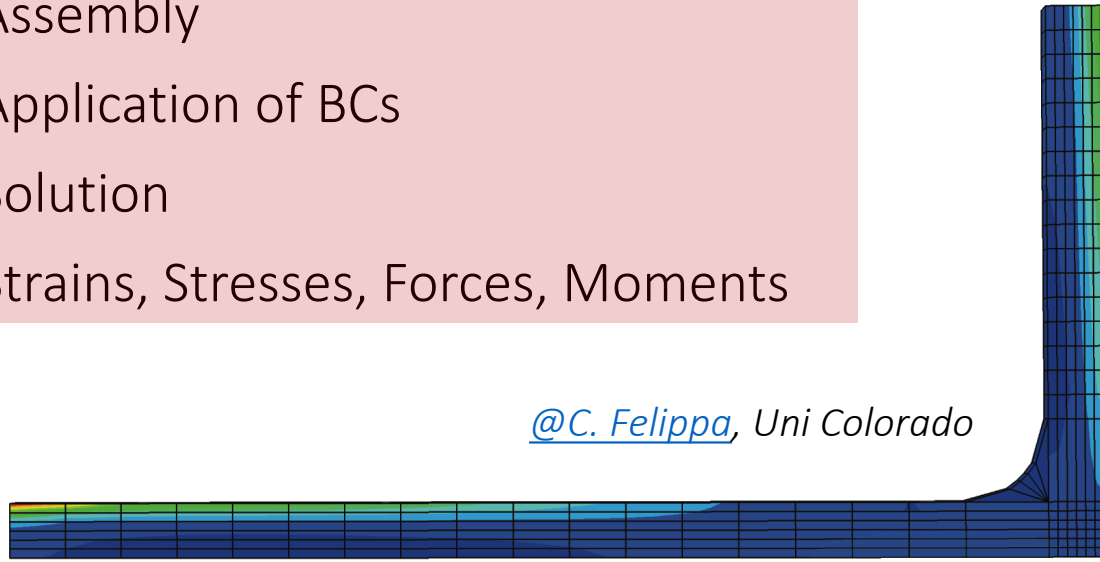
Assembly

Application of BCs

Solution

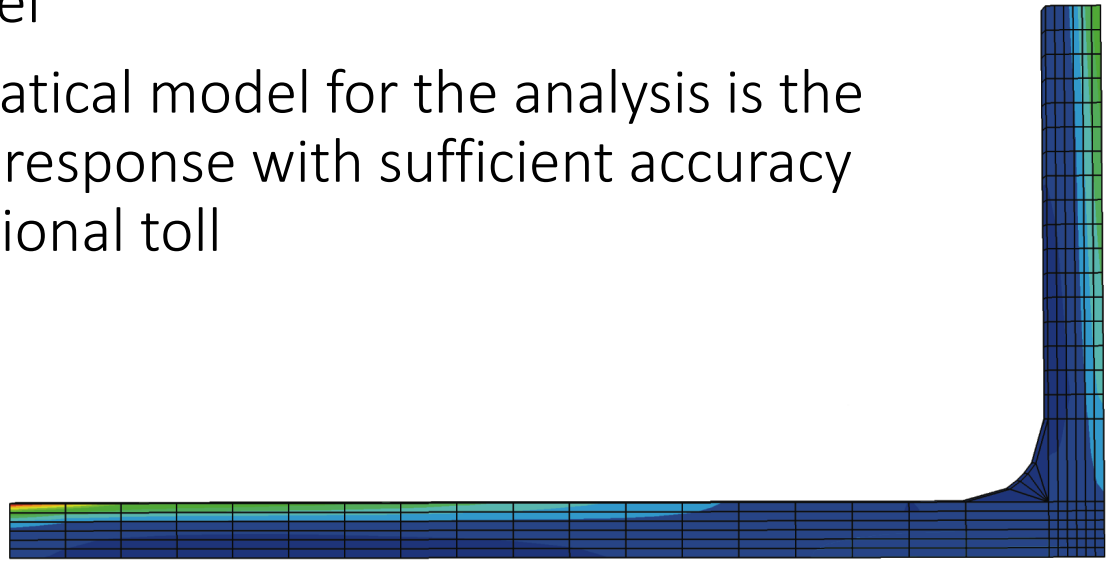
Strains, Stresses, Forces, Moments

[@C. Felippa](#), Uni Colorado

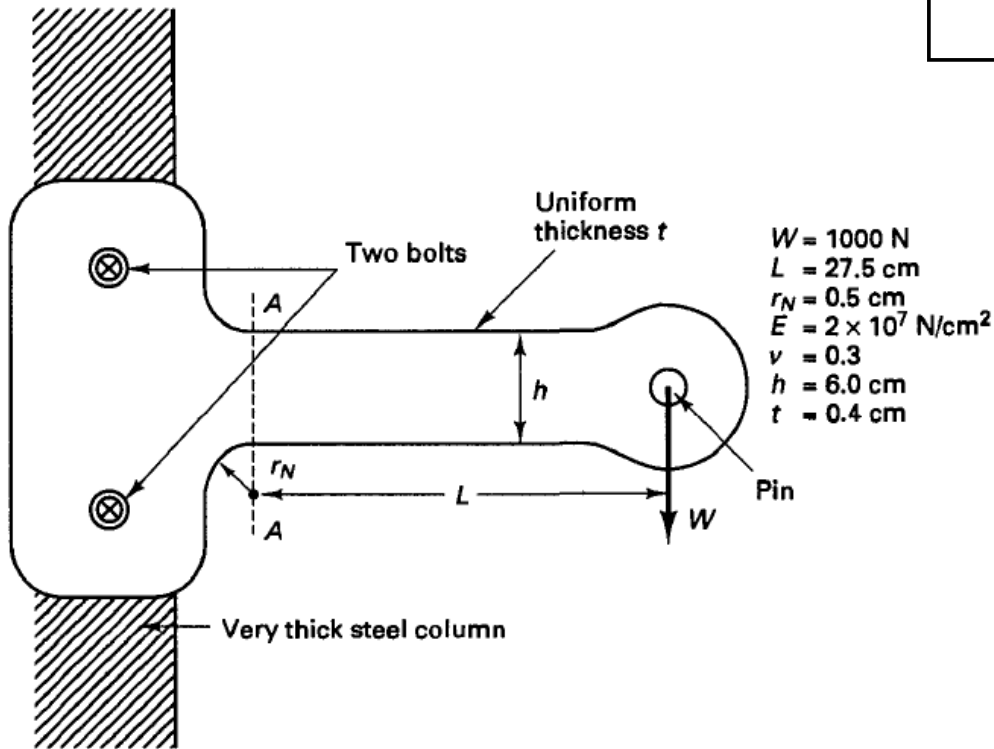


Challenges

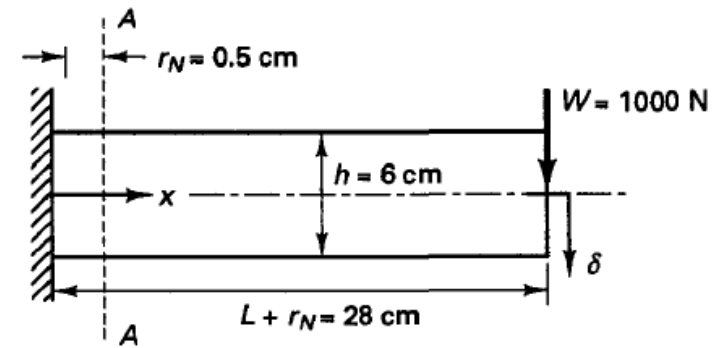
- The MFE is only a way of solving the mathematical model
- The solution of the physical problem depends on the quality of the mathematical model – the choice of the mathematical model is crucial
- The chosen mathematical model is reliable if the required response can be predicted within a given level of accuracy compared to the response of a very comprehensive (highly refined) mathematical model
- The most effective mathematical model for the analysis is the one that gives the required response with sufficient accuracy and at the lowest computational toll



Simple Example



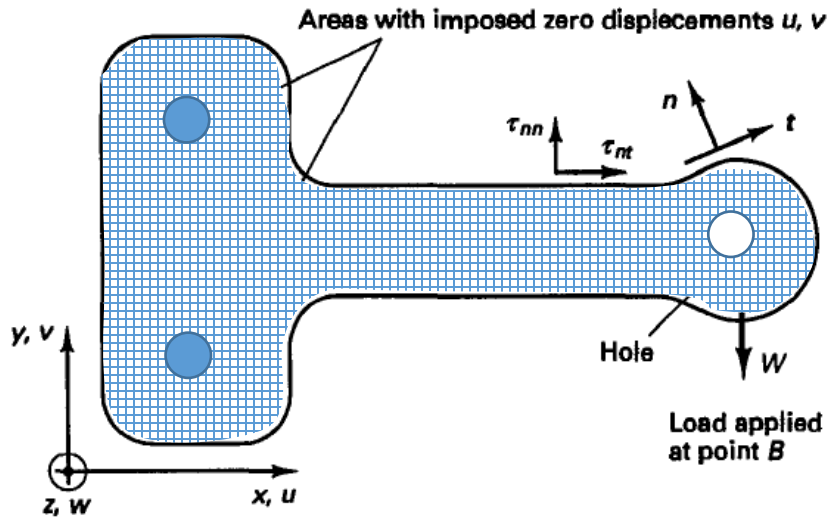
$$k_{loc} = \frac{EI}{L^3} \begin{bmatrix} \frac{AL^2}{I} & 0 & 0 & -\frac{AL^2}{I} & 0 & 0 \\ 0 & 3 & 3L & 0 & -3 & 0 \\ 0 & 3L & 3L^2 & 0 & -3L & 0 \\ -\frac{AL^2}{I} & 0 & 0 & \frac{AL^2}{I} & 0 & 0 \\ 0 & -3 & -3L & 0 & 3 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$



$$\begin{aligned}
 M &= WL \\
 &= 27,500 \text{ N cm} \\
 \delta|_{\text{at load } W} &= \frac{1}{3} \frac{W(L + r_N)^3}{EI} + \frac{W(L + r_N)}{\frac{5}{6}AG} \\
 &= 0.053 \text{ cm}
 \end{aligned}$$

Complex physical problem modelled by a **simple mathematical model**

Simple Example



$$\delta|_{\text{at load } W} = 0.064 \text{ cm}$$

$$M|_{x=0} = 27,500 \text{ N cm}$$

Equilibrium equations

$$\left. \begin{aligned} \frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} &= 0 \\ \frac{\partial \tau_{yx}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} &= 0 \end{aligned} \right\} \text{ in domain of bracket}$$

$\tau_{nn} = 0, \tau_{nt} = 0$ on surfaces except at point B and at imposed zero displacements

Stress-strain relation

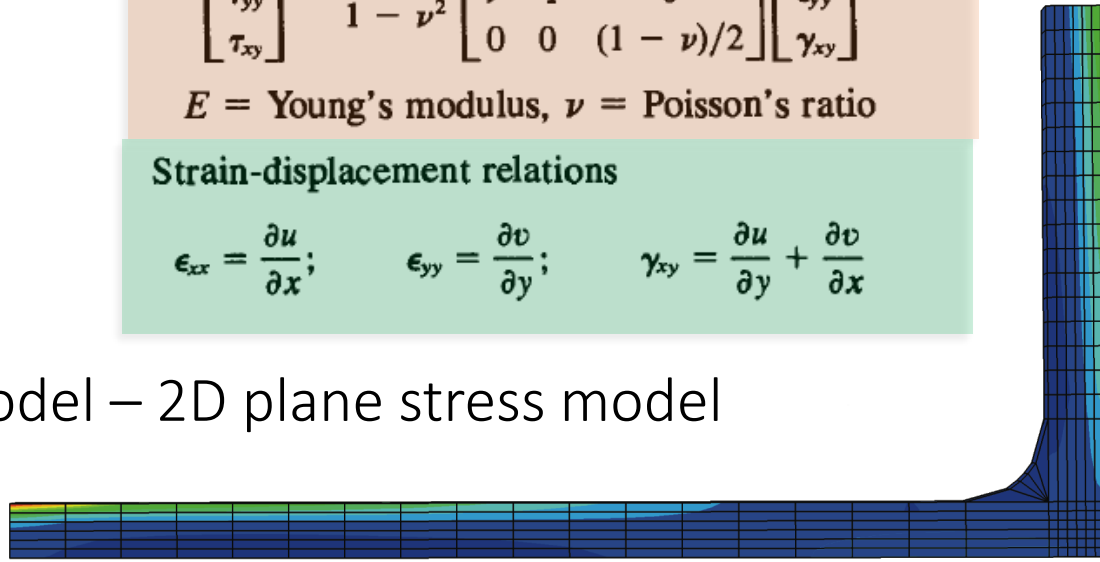
$$\begin{bmatrix} \tau_{xx} \\ \tau_{yy} \\ \tau_{xy} \end{bmatrix} = \frac{E}{1 - \nu^2} \begin{bmatrix} 1 & \nu & 0 \\ \nu & 1 & 0 \\ 0 & 0 & (1 - \nu)/2 \end{bmatrix} \begin{bmatrix} \epsilon_{xx} \\ \epsilon_{yy} \\ \gamma_{xy} \end{bmatrix}$$

E = Young's modulus, ν = Poisson's ratio

Strain-displacement relations

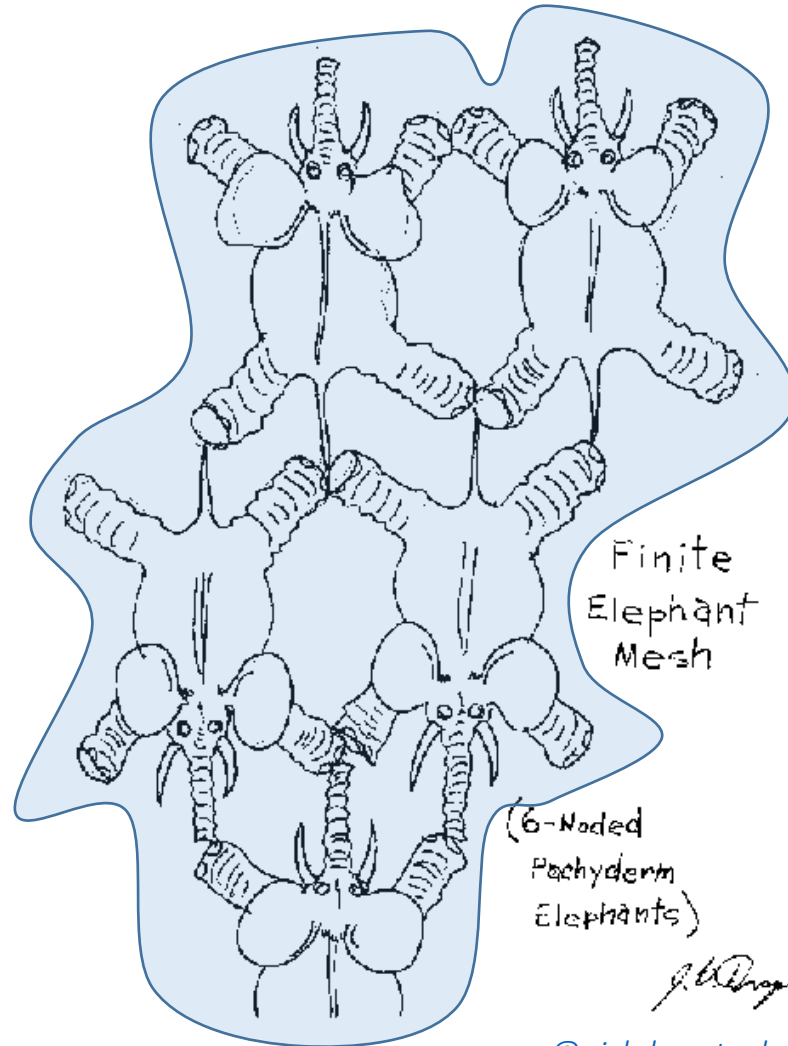
$$\epsilon_{xx} = \frac{\partial u}{\partial x}; \quad \epsilon_{yy} = \frac{\partial v}{\partial y}; \quad \gamma_{xy} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$$

Detailed reference model – 2D plane stress model



Discretization of structures into finite elements

How to discretize
the continuum?



[@eislab.gatech](https://twitter.com/eislab.gatech)

Discretization of structures into finite elements

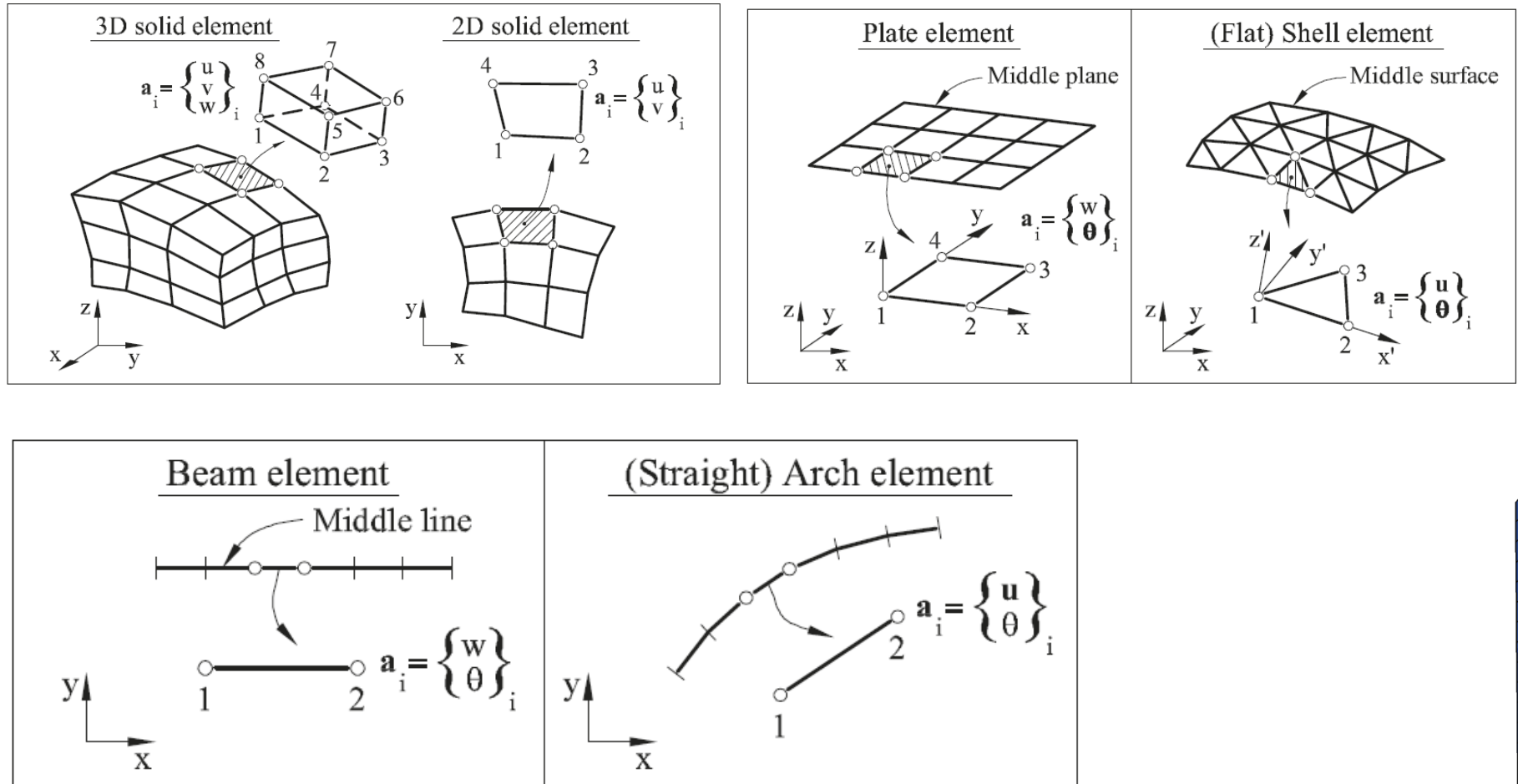
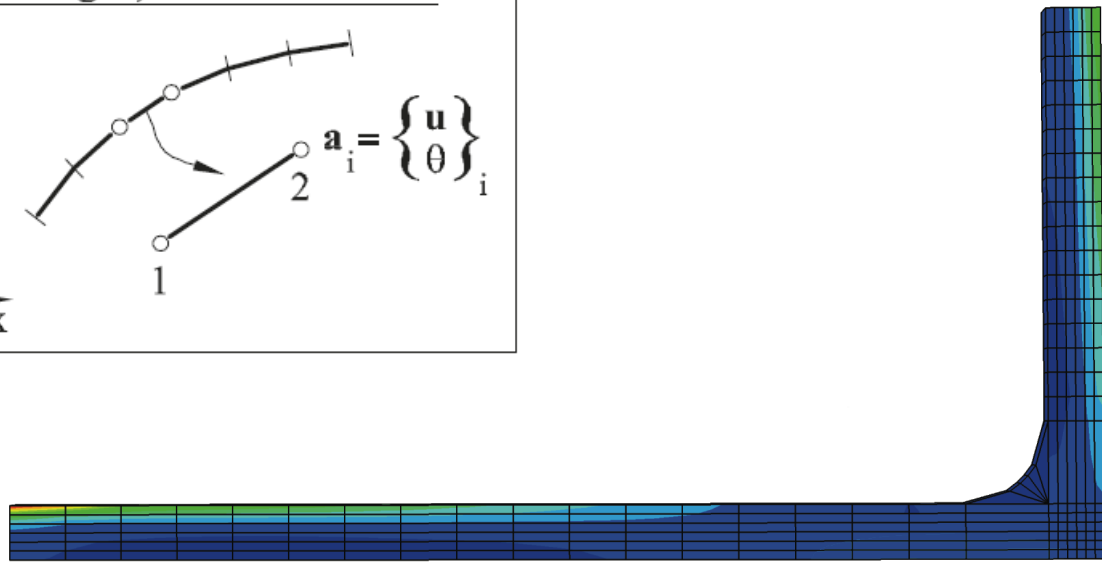
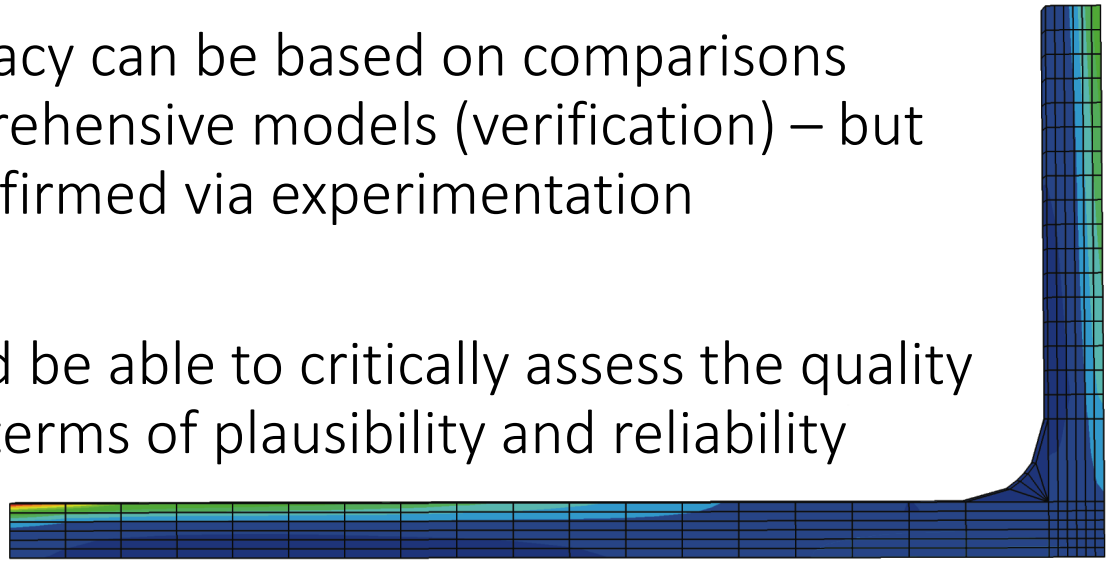


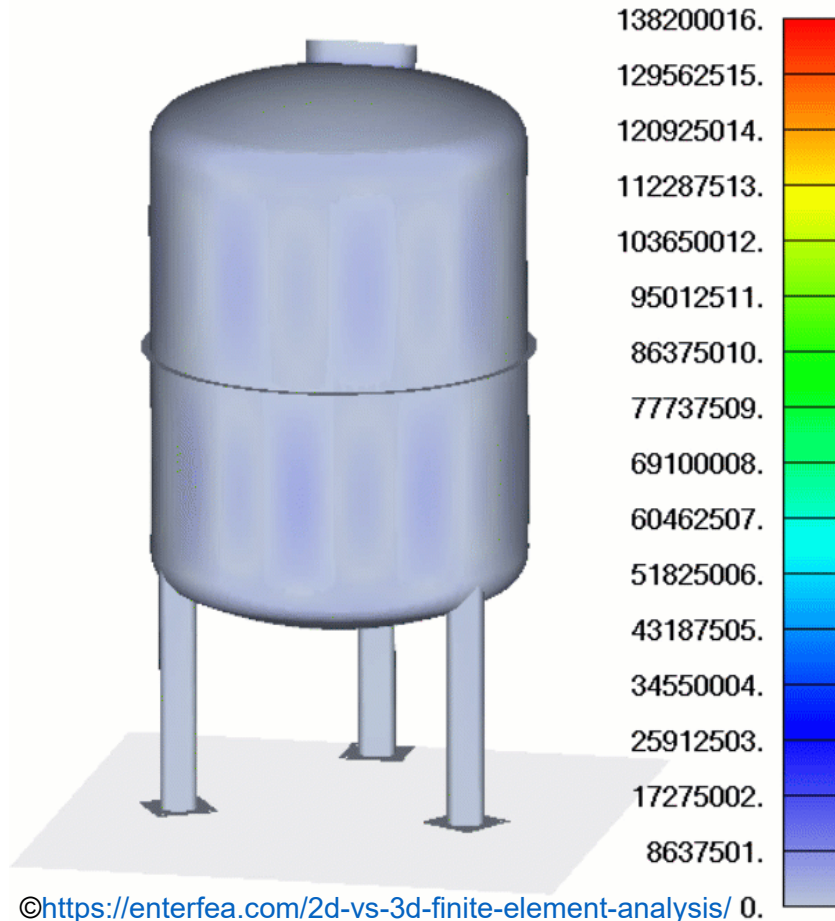
Fig 1.4 from Onate, Vol. 1



What to keep in Mind

- Choice of mathematical model must correspond to the desired simulation objective
- The most effective mathematical model delivers reliable answers with the least amount of effort
- Any solution (including MFE) of a mathematical model is limited to information contained in or fed into the model:
bad input – bad output (garbage in – garbage out)
- V&V: Assessment of accuracy can be based on comparisons against results from comprehensive models (verification) – but in practice it has to be confirmed via experimentation (validation)
- The engineer (user) should be able to critically assess the quality of the obtained results in terms of plausibility and reliability





©<https://enterfea.com/2d-vs-3d-finite-element-analysis/> 0.

Let's start modelling in FEM!

